Cloudian HyperIQ
User's Guide

Version 1.9

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What's New in HyperIQ Version 1.9

The main new features and enhancements in the Cloudian HyperIQ 1.9 release are:

- Alert codes -- Each type of alert in HyperIQ now has a unique alphanumeric alert code. These codes make it easier to identify alerts and to communicate with Cloudian Support about alerts. For alerts that may appear both in HyperIQ and in HyperStore's Cloudian Management Console (CMC), the same alert code is used by both applications. For more information see **"Alert Codes"** (page 45).
- SMART status alerts for disk drives -- HyperIQ now generates an alert if the SMART (Self-Monitoring, Analysis, and Reporting Technology) status of an HDD or SSD in your HyperStore system reports a status other than "OK".
- Support for monitoring HyperStore "auxiliary nodes" -- If in your HyperStore system you've added "auxiliary nodes" that run the HyperStore File Service or the HyperStore Search Service (available starting in HyperStore 7.5.1), HyperIQ now supports basic monitoring of those nodes. If you are setting up a new HyperIQ instance, you can execute the installation in a way that includes monitoring of HyperStore auxiliary nodes (see **"Installing or Upgrading HyperIQ"** (page 9), particularly the installation command option *--auxiliary-nodes*). If you have an existing HyperIQ instance monitoring an existing HyperStore system to which you've recently added the HyperStore File Service and/or the HyperStore Search Service, you can modify your HyperIQ configuration to include monitoring of the HyperStore auxiliary nodes (see **"Using the iqmgr Tool to Manage HyperIQ"** (page 27) and particularly **"Updating HyperIQ to Monitor Additional HyperStore Resources"** (page 27)).
- Support for High Availability (HA) -- If you wish you can now deploy two separate instances of HyperIQ to monitor one HyperStore system, so that if one HyperIQ instance becomes unavailable the other HyperIQ instance can still be used. The mechanism by which HyperIQ collects HyperStore statistics has been revised so that statistics can be efficiently pushed from HyperStore nodes to one or multiple HyperIQ instances. For more information see **"Installing or Upgrading HyperIQ"** (page 9), particularly the installation command option *--secondary-hiq* (which you will use when installing a second HyperIQ instance to monitor a HyperStore system that is already being monitored by another HyperIQ instance).

Note For more granular release details including bug fixes please see the release notes.

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Chapter 1. Installing or Upgrading HyperIQ

Cloudian HyperIQ™ is a solution for monitoring the health and performance of a Cloudian HyperStore system. HyperIQ provides monitoring data visualization and analytics as well as multi-channel alerting. This document describes how to install HyperIQ version 1.9, connect it to your HyperStore system, and start using it.

This chapter covers the following topics:

- **"Requirements"** (page 9)
- **"Installing or Upgrading HyperIQ as a Virtual Appliance"** (page 10)
- **"Installing or Upgrading HyperIQ in an Existing Docker Environment"** (page 14)
- **"Using an Existing Grafana Deployment to Display HyperIQ Data"** (page 17)

1.1. Requirements

The table below describes what you will need to install and use HyperIQ.

Requirement	Description
Virtualization or containerization platform	HyperIQ is available in two forms: • A virtual appliance, for installation on VMware (ESXi 5.5 or newer, or Workstation 10.x or newer) or VirtualBox. • A containerized application, for installation on Docker (requires docker 19.03 or newer **and** docker-compose 1.17 or newer; Kubernetes is not currently supported)
HyperIQ OVA or installer	From the Cloudian Support portal, download one of the following: • *hyperiq-1.9.ova*, for installing HyperIQ as a virtual appliance • *hyperiq-1.9.bin*, for upgrading an existing HyperIQ virtual appliance; or for installing or upgrading HyperIQ in an existing Docker environment on Linux
HyperStore system	HyperIQ has been verified to work with HyperStore 7.1 and newer. One HyperIQ instance monitors one or more entire HyperStore systems.
HyperStore license that supports HyperIQ (only required for S3 Analytics)	To view S3 Analytics data in HyperIQ, your HyperStore system license must include HyperIQ Enterprise support. To view other types of monitoring data in HyperIQ -- such as data pertaining to HyperStore system health and performance -- no special license is required (any HyperStore license supports using HyperIQ for this health data).
Network access between HyperIQ and HyperStore	Your network must allow: • Each HyperIQ host to access each HyperStore host on ports 22 and 9999 • Each HyperStore host to access each HyperIQ host on port 3443 • If your HyperStore system includes HyperStore File Service, HyperStore Search Service, and/or HyperBalance hosts, HyperIQ must be able to access those hosts on ports 22 and 9999. Those hosts do not need to

Requirement	Description
	access HyperIQ on port 3443 or any other port. **Note** If your HyperStore version is 7.2.0 or 7.2.1 you must disable the HyperStore Firewall if you want to use HyperIQ. For instructions, in your HyperStore 7.2.x documentation see section 4.3.2 "HyperStore Firewall". However, do not disable the HyperStore Firewall if your HyperStore 7.2.0 or 7.2.1 deployment lacks a dedicated internal network for HyperStore nodes. Instead, wait until you've upgraded to HyperStore 7.2.2 or newer before using HyperIQ. In HyperStore 7.2.2 and newer the HyperStore Firewall allows HIQ access by default.

1.2. Installing or Upgrading HyperIQ as a Virtual Appliance

Note The instructions immediately below are for doing a fresh installation of HyperIQ as a virtual appliance. For instructions to upgrade an existing HyperIQ virtual appliance, jump ahead to **"Upgrading an Existing HyperIQ Virtual Appliance to Version 1.9"** (page 14).

Note If you are using the HyperBalance load balancer with your HyperStore system and **if you want HyperIQ to monitor HyperBalance** (as well as monitoring your HyperStore system), before installing HyperIQ you must enable the Prometheus exporter on each of your HyperBalance nodes. On each HyperBalance node, log into the HyperBalance UI, go to **Cluster Configuration -> Layer 7 -- Advanced Configuration** and then check the "Enable Prometheus Exporter" checkbox.

1.2.1. Deploying the HyperIQ VM

Use your virtualization platform console to deploy a new virtual machine (VM) from the HyperIQ OVA file that you downloaded from the Cloudian Support portal.

For example, for VMware ESXi:

1. From the device on which you downloaded the HyperIQ OVA file, connect to your ESXi Server console and log in.
2. On the left side of the ESXi console, select **Virtual Machines**.
3. Click **Create / Register VM**. This launches the **New virtual machine** creation wizard.
4. For the VM creation type select “Deploy a virtual machine from an OVF or OVA file”, then click **Next**.
5. Enter your desired name for the virtual machine, then click in the file selection area of the wizard screen and browse (on your local device) to the HyperIQ OVA file that you downloaded. After selecting the OVA file, click **Next**.
6. In the remaining screens of the VM creation wizard, accept the default values (or make adjustments if you wish).
7. In the wizard's **Ready to complete** screen, review the attributes of the new VM and then click **Finish**.

   Although the wizard will close and the new VM will appear in the **Virtual machines** list right away, it will

take some time -- about 10-15 minutes depending on your environment -- for creation of the VM to complete. You can track the progress by opening the **Recent tasks** panel at the bottom of the **Virtual machines** list.

The HyperIQ VM that is created has these specs:

- 2 CPU cores, 8GB RAM, one 150GB disk for the OS and data
- OS = Ubuntu 18.04 Linux 64-bit
- hostname = Defaults to *hyperiq* but is configurable during HyperIQ setup (as described further below)

IMPORTANT ! The HyperIQ VM stores its data locally, on the 150GB disk, without any replication. In a production environment you should take appropriate precautions to ensure data durability, such as performing regular backups of the VM.

1.2.2. Integrating the HyperIQ VM with HyperStore

If the HyperIQ VM is not already powered on, power it on now. Then **log into the VM's console** with these credentials:

- User name: **cloudian**
- Password: **Cloudian1!**

You will be prompted to create a new password for the 'cloudian' user. The change password prompt will require you first to enter again your "current UNIX password" -- which is Cloudian1! -- and then your desired new password. Remember the new password, since you will need it whenever you want to log into the HyperIQ virtual machine. (Note that this password for logging into the HyperIQ host is a different password than the password for logging into the HyperIQ GUI. You will set a HyperIQ GUI password later in this installation procedure.)

Once you're logged in, a HyperIQ installer tool will lead you through the process of integrating HyperIQ with your HyperStore cluster. The command for launching the installer is:

```
$ iqinstall [--replace-ips 'PRIVATE_IP1=PUBLIC_IP1,PRIVATE_IP2=PUBLIC_IP2...']
[-k FULL_PATH_TO_KEY_FILE] [--auxiliary-nodes] [--with-managed-services] [--secondary-hiq]
```

Options:

- *--replace-ips 'PRIVATE_IP1=PUBLIC_IP1,PRIVATE_IP2=PUBLIC_IP2...'*

  You must use this option if in your HyperStore system configuration you configured your installation "survey file" with IP addresses from a private back-end network (which is an atypical survey file configuration) and your HyperIQ node cannot access that private network. With this option you specify, for each HyperStore node, the private IP address and its corresponding public IP address (which HyperIQ can access). This example is for a three-node HyperStore system:

```
$ iqinstall --replace-ips '10.10.210.2=10.10.1.212,10.10.210.3=10.10.1.213,10.10.210.4=10.10.1.214'
```

- *-k FULL_PATH_TO_KEY_FILE*

  If you want HyperIQ to use an SSH key that you have already placed on the HyperIQ host to make SSH connections to HyperStore nodes -- rather than using an SSH password that you can specify during the install process as described further below -- launch the installer with the *-k* option. For example you can copy the SSH key from your HyperStore Configuration Master node (in the installation staging directory) over to your HyperIQ node, and then specify that file with the the *-k* option.

If you have multiple SSH keys -- such as if you are having HyperIQ monitor multiple HyperStore clusters and you have different SSH keys for the different clusters; or if you are having HyperIQ monitor a HyperStore cluster that includes auxiliary nodes (HyperStore File and/or HyperStore Search nodes) and you have a different key for the auxiliary nodes than for the rest of your HyperStore nodes -- you can specify multiple *-k* flags each with a different key, like *-k PATH_TO_KEY_FILE1 -k PATH_TO_KEY_FILE2*.

> **Note** The option of using an SSH key rather than a password is supported only for *root* user access to HyperStore nodes. For accessing HyperStore as the *sa_admin* user (in the scenario where you've enabled the HyperStore Shell and disabled root access to your HyperStore nodes), only password authentication is supported currently.

- *--auxiliary-nodes*

  Include this option if your HyperStore system includes HyperStore File nodes and/or HyperStore Search nodes and you want HyperIQ to monitor those nodes (as well as monitoring the rest of your HyperStore nodes).

- *--with-managed-services*

  Include this option only if instructed to do so by Cloudian Support. This will be applicable only if you are a Support Plus, Premier Support, or HyperCare customer, and you have already received and installed (in your HyperStore system) the special HyperStore license file associated with these Support programs.

  --secondary-hiq

  Include this option if this is a secondary HyperIQ instance, monitoring a HyperStore system that is also being monitored by another HyperIQ instance that you have already installed. This option facilitates a high availability (HA) HyperIQ deployment, whereby two separate HyperIQ instances monitor the same HyperStore system so that if one HyperIQ instance becomes unavailable the other HyperIQ instance can still be used. Note that although this installation option refers to a "secondary" HyperIQ, in fact both HyperIQ instances will be equal in their role and operation -- neither instance will be in a "primary" role and neither instance will be in a back-up or fail-over role. Put differently, the two HyperIQ instances will be deployed in an "Active / Active" configuration. (The only difference is in regard to "manage services" configuration [if applicable] and certain aspects of metrics log processing; having just one of the two HyperIQ instances be installed as "*--secondary-hiq*" avoids an undesirable duplication of these tasks.)

When you launch the installer you will be first prompted to **configure basic networking settings** for the HyperIQ node:

- Hostname (defaults to *hyperiq*)
- Networking mode (DHCP or static)
- IP address
- Default gateway
- DNS host(s)

At most prompts a default value is displayed, which you can accept by pressing Enter if you wish.

Once the installer applies the network settings, it sets up the core HyperIQ components as Docker containers. This includes containerized instances of the mature open source technologies **Prometheus** and **Grafana**, which HyperIQ leverages for the collection, analysis, and visualization of HyperStore monitoring data. This automated setup process may take a few minutes.

Next will be a series of prompts for information that HyperIQ needs to integrate with your HyperStore cluster. This includes:

- A **unique name for your HyperStore cluster**. Only lower case letters, dashes, and underscores are allowed. Here, "cluster" means an entire HyperStore system (which depending on your deployment topology, may encompass multiple data centers or regions). The name that you give to your cluster is only for purposes of organizing data within HyperIQ and does not need to match any name that you've configured within HyperStore. Note that if you are connecting HyperIQ to multiple HyperStore clusters (multiple independent HyperStore systems that you want to monitor), enter a name for one of the clusters here, and then later in the installation process -- after HyperIQ has been successfully integrated with the first cluster-- you will be given the opportunity to integrate HyperIQ with an additional cluster or clusters. If you do integrate HyperIQ with multiple HyperStore clusters you will need to give each HyperStore cluster a different name.

  > **Note** If this is a secondary HyperIQ instance -- monitoring a HyperStore cluster that is also being monitored by another HyperIQ instance that you've already installed -- use the same HyperStore cluster name that you did when you installed the first HyperIQ instance. The installer does not enforce this as a requirement, but it will be more logical for HyperIQ users if the HyperStore cluster has the same name in the UIs of both HyperIQ instances.

- the **IP address of one of your HyperStore nodes** within the cluster (it can be any node, and the IP address must be reachable by HyperIQ), and the user name (either *root* or *sa_admin*) and password for connecting to that node. Note that the password prompt will not appear if you're using an SSH key instead. From the one HyperStore IP address that you provide, the HyperIQ installer will automatically obtain the IP addresses of all of the rest of your HyperStore nodes. The installer then installs Prometheus "exporters" on every HyperStore node. After this step completes, the installer finalizes the configuration of Prometheus and Grafana on the HyperIQ VM.
- The **IP address of your HyperBalance load balancer**, if you are using HyperBalance with your HyperStore cluster and you would like HyperBalance statistics to appear in HyperIQ. (See the Note about HyperBalance at the beginning of **"Installing or Upgrading HyperIQ as a Virtual Appliance"** (page 10).)

Lastly, the installer prompts you to specify a password for logging into the HyperIQ UI. If you wish you can use special characters in the password as well as alphanumeric characters. Remember this password, since you will need it whenever you want to access HyperIQ UI.

After the installer completes it returns you to the terminal prompt. Next you can use a browser to access the HyperIQ UI as described in **"Getting Started With HyperIQ"** (page 19).

> **Note** If you have a very large HyperStore cluster it's possible that you may encounter **copy timeout errors** (displayed in your terminal) as the HyperIQ set-up process tries to concurrently copy package files to multiple HyperStore nodes through your network. In that case, run the HyperIQ installer again but this time add the option *-t 5* (for example *iqinstall -t 5*). This reduces the number of concurrent threads used during HyperIQ set-up (so that 5 threads are used rather than the default of 10) which will make the aggregate set-up time somewhat longer but should eliminate the copy time-out problem.

Note For its access to port 9999 on your HyperStore nodes, HyperIQ uses HTTPS (TLS v. 1.2) with Basic Authentication. The HyperIQ setup tool automatically configures these security mechanisms using a self-signed certificate for TLS and a randomly generated password for Basic Authentication.

1.2.3. Upgrading an Existing HyperIQ Virtual Appliance to Version 1.9

To upgrade an existing HyperIQ virtual appliance to version 1.9, download the *hyperiq-1.9.bin* file (**not** the *.ova* file) from the Cloudian Support portal, put it into a working directory on your HyperIQ virtual appliance, and run these commands:

```
$ chmod +x hyperiq-1.9.bin
$ ./hyperiq-1.9.bin install [--replace-ips 'PRIVATE_IP1=PUBLIC_IP1,PRIVATE_IP2=PUBLIC_IP2...']
[-k KEY_FILENAME] [--auxiliary-nodes] [--with-managed-services] [--secondary-hiq]
```

Follow the prompts to complete the upgrade.

For descriptions of the command options, see the **"Integrating the HyperIQ VM with HyperStore"** (page 11) section.

1.3. Installing or Upgrading HyperIQ in an Existing Docker Environment

Note If you previously installed HyperIQ in an existing Docker environment and now you want to **upgrade** to HyperIQ version 1.9, follow the steps in this section. The upgrade procedure is the same as the procedure for doing a fresh installation.

Note If you are using the HyperBalance load balancer with your HyperStore system and **if you want HyperIQ to monitor HyperBalance** (as well as monitoring your HyperStore system), before installing HyperIQ you must enable the Prometheus exporter on each of your HyperBalance nodes. On each HyperBalance node, log into the HyperBalance UI, go to **Cluster Configuration -> Layer 7 -- Advanced Configuration** and then check the "Enable Prometheus Exporter" checkbox.

If the Docker environment in which you are installing HyperIQ is a VM dedicated solely to HyperIQ, the following VM configuration is recommended:

- 2 CPU cores, 8GB RAM, one 150GB disk
- OS = Ubuntu 18.04 (or 20.04) Linux 64-bit

If you are installing HyperIQ in a shared Docker environment, it is recommended that 2 CPU cores, 8GB RAM, and 150GB disk space be available to HyperIQ.

To install HyperIQ version 1.9 in a Docker environment:

1. On the host machine on which you want to install HyperIQ, log in as a user who has permissions to run *docker* commands. In a working directory on the host machine, download the HyperIQ installer (*hyperiq-1.9.bin*) from the Cloudian Support portal.

   Note You must use a command terminal for the remainder of this procedure.

2. Change into the working directory in which you've downloaded the HyperIQ installer file, if you're not already in that directory.
3. Run the following command to make the installer file executable:

```
chmod +x hyperiq-1.9.bin
```

4. Start the installation:

```
./hyperiq-1.9.bin install [--replace-ips 'PRIVATE_IP1=PUBLIC_IP1,PRIVATE_IP2=PUBLIC_IP2...']
[-k FULL_PATH_TO_KEY_FILE] [--auxiliary-nodes] [--with-managed-services] [--secondary-hiq]
```

Options:

- *--replace-ips 'PRIVATE_IP1=PUBLIC_IP1,PRIVATE_IP2=PUBLIC_IP2...'*

  You must use this option if in your HyperStore system configuration you configured your installation "survey file" with IP addresses from a private back-end network (which is an atypical survey file configuration) and your HyperIQ node cannot access that private network. With this option you specify, for each HyperStore node, the private IP address and its corresponding public IP address (which HyperIQ can access). This example is for a three-node HyperStore system:

```
$ ./hyperiq-1.9.bin install --replace-ips '10.10.210.2=10.10.1.212,10.10.210.3=10.10.1.213,
10.10.210.4=10.10.1.214'
```

- *-k FULL_PATH_TO_KEY_FILE*

  If you want HyperIQ to use an SSH key that you have already placed on the HyperIQ host to make SSH connections to HyperStore nodes -- rather than using an SSH password that you can specify during the install process as described further below -- launch the installer with the *-k* option. For example you can copy the SSH key from your HyperStore Configuration Master node (in the installation staging directory) over to your HyperIQ node, and then specify that file with the the *-k* option.

  If you have multiple SSH keys -- such as if you are having HyperIQ monitor multiple HyperStore clusters and you have different SSH keys for the different clusters; or if you are having HyperIQ monitor a HyperStore cluster that includes auxiliary nodes (HyperStore File and/or HyperStore Search nodes) and you have a different key for the auxiliary nodes than for the rest of your HyperStore nodes -- you can specify multiple *-k* flags each with a different key, like *-k PATH_TO_KEY_FILE1 -k PATH_TO_KEY_FILE2*.

  > **Note** The option of using an SSH key rather than a password is supported only for *root* user access to HyperStore nodes. For accessing HyperStore as the *sa_admin* user (in the scenario where you've enabled the HyperStore Shell and disabled root access to your HyperStore nodes), only password authentication is supported currently.

- *--auxiliary-nodes*

  Include this option if your HyperStore system includes HyperStore File nodes and/or HyperStore Search nodes and you want HyperIQ to monitor those nodes (as well as monitoring the rest of your HyperStore nodes)

- *--with-managed-services*

  Include this option only if instructed to do so by Cloudian Support. This will be applicable only if you are a Support Plus, Premier Support, or HyperCare customer, and you have already received and installed (in your HyperStore system) the special HyperStore license file associated with these Support programs.

  --secondary-hiq

  Include this option if this is a secondary HyperIQ instance, monitoring a HyperStore system that is also

being monitored by another HyperIQ instance that you have already installed. This option facilitates a high availability (HA) HyperIQ deployment, whereby two separate HyperIQ instances monitor the same HyperStore system so that if one HyperIQ instance becomes unavailable the other HyperIQ instance can still be used. Note that although this installation option refers to a "secondary" HyperIQ, in fact both HyperIQ instances will be equal in their role and operation -- neither instance will be in a "primary" role and neither instance will be in a back-up or fail-over role. Put differently, the two HyperIQ instances will be deployed in an "Active / Active" configuration. (The only difference is in regard to "managed services" configuration [if applicable] and certain aspects of metrics log processing; having just one of the two HyperIQ instances be installed as "*--secondary-hiq*" avoids an undesirable duplication of these tasks.)

The installer sets up the core HyperIQ components as Docker containers. This includes containerized instances of the mature open source technologies **Prometheus** and **Grafana**, which HyperIQ leverages for the collection, analysis, and visualization of HyperStore monitoring data. This automated setup process may take a few minutes.

Next will be a series of prompts for information that HyperIQ needs to integrate with your HyperStore cluster. This includes:

- A **unique name for your HyperStore cluster**. Only lower case letters, dashes, and underscores are allowed. Here, "cluster" means an entire HyperStore system (which depending on your deployment topology, may encompass multiple data centers or regions). The name that you give to your cluster is only for purposes of organizing data within HyperIQ and does not need to match any name that you've configured within HyperStore. Note that if you are connecting HyperIQ to multiple HyperStore clusters (multiple independent HyperStore systems that you want to monitor), enter a name for one of the clusters here, and then later in the installation process -- after HyperIQ has been successfully integrated with the first cluster-- you will be given the opportunity to integrate HyperIQ with an additional cluster or clusters. If you do integrate HyperIQ with multiple HyperStore clusters you will need to give each HyperStore cluster a different name.

  > **Note** If this is a secondary HyperIQ instance -- monitoring a HyperStore cluster that is also being monitored by another HyperIQ instance that you've already installed -- use the same HyperStore cluster name that you did when you installed the first HyperIQ instance. The installer does not enforce this as a requirement, but it will be more logical for HyperIQ users if the HyperStore cluster has the same name in the UIs of both HyperIQ instances.

- The **IP address of one of your HyperStore nodes** within the cluster (it can be any node, and the IP address must be reachable by HyperIQ), and the user name (either *root* or *sa_admin*) and password for connecting to that node. Note that the password prompt will not appear if you're using an SSH key instead. From the one HyperStore IP address that you provide, the HyperIQ installer will automatically obtain the IP addresses of all of the rest of your HyperStore nodes. The installer then installs Prometheus "exporters" on every HyperStore node. After this step completes, the installer finalizes the configuration of Prometheus and Grafana on the HyperIQ VM.
- The **IP address of your HyperBalance load balancer**, if you are using HyperBalance with your HyperStore cluster and you would like HyperBalance statistics to appear in HyperIQ. (See the Note about HyperBalance at the beginning of **"Installing or Upgrading HyperIQ in an Existing Docker Environment"** (page 14).)

Lastly, the installer prompts you to specify a password for logging into the HyperIQ UI. If you wish you can use special characters in the password as well as alphanumeric characters. Remember this password, since you will need it whenever you want to access HyperIQ UI.

After the installer completes it returns you to the terminal prompt. Next you can use a browser to access the HyperIQ UI as described in **"Getting Started With HyperIQ"** (page 19).

Leave the (*hyperiq-1.9.{bin,app,exe}* file in the working directory -- you will need it if at some point in the future you want to uninstall HyperIQ.

Note If you have a very large HyperStore cluster it's possible that you may encounter **copy timeout errors** (displayed in your terminal) as the HyperIQ set-up process tries to concurrently copy package files to multiple HyperStore nodes through your network. In that case, run the HyperIQ installer again but this time add the option *-t 5* (for example *./hyperiq-1.9.bin install -t 5*). This reduces the number of concurrent threads used during HyperIQ set-up (so that 5 threads are used rather than the default of 10) which will make the aggregate set-up time somewhat longer but should eliminate the copy time-out problem.

Note For its access to port 9999 on your HyperStore nodes, HyperIQ uses HTTPS (TLS v. 1.2) with Basic Authentication. The HyperIQ installer automatically configures these security mechanisms, using a self-signed certificate for TLS and a randomly generated password for Basic Authentication.

1.4. Using an Existing Grafana Deployment to Display HyperIQ Data

1. Install HyperIQ and integrate it with your HyperStore cluster, as described in **"Installing or Upgrading HyperIQ as a Virtual Appliance"** (page 10) or **"Installing or Upgrading HyperIQ in an Existing Docker Environment"** (page 14).
2. In your existing Grafana UI, go to **Configuration -> Data Sources**. Click **Add data source**.
3. Select Prometheus as the data source type. When configuring the data source:
   - The data source name must be *Prometheus*
   - The data source URL by default is *http://<HyperIQ FQDN or IP address>:3000/prometheus*
   - Basic Auth must be enabled, with user name *cloudian_admin* and the Prometheus/Alertmanager UI Basic Auth password that you set during HyperIQ set-up.
4. After Saving the data source configuration, select the **Dashboards** tab for the new data source and import the HyperIQ dashboards.

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Chapter 2. Getting Started With HyperIQ

This chapter covers the following topics:

- **"Connecting to the HyperIQ UI"** (page 19)
- **"Viewing HyperIQ Dashboards and Panels"** (page 19)
- **"Adding HyperIQ Users"** (page 24)

2.1. Connecting to the HyperIQ UI

To connect to the HyperIQ UI point a browser to either of the following URLs:

http://<HyperIQ IP address>:3000

https://<HyperIQ IP address>:3443 (this uses a self-signed TLS certificate)

HyperIQ comes with two pre-configured users:

- **cloudian_admin** -- This user has full privileges including the ability to view and edit dashboards, and to create new HyperIQ users. The password for this user is the HyperIQ UI password that you specified during HyperIQ set-up.
- **org_editor** -- This user has the ability to view and edit dashboards, but not to create new users. The default password for this user is *admin*. When logging in as this user you will be prompted to change the user's password, which you can do within the flow of the login process.

If you are logging in shortly after completing the installation of HyperIQ, note that it may take several minutes for HyperIQ to become populated with HyperStore monitoring data.

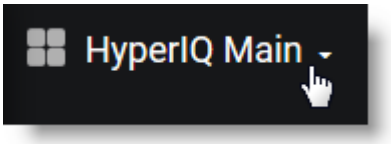
2.2. Viewing HyperIQ Dashboards and Panels

When you've logged in to HyperIQ for the first time you are at the **HyperIQ Main** dashboard. This is one of several HyperIQ dashboards:

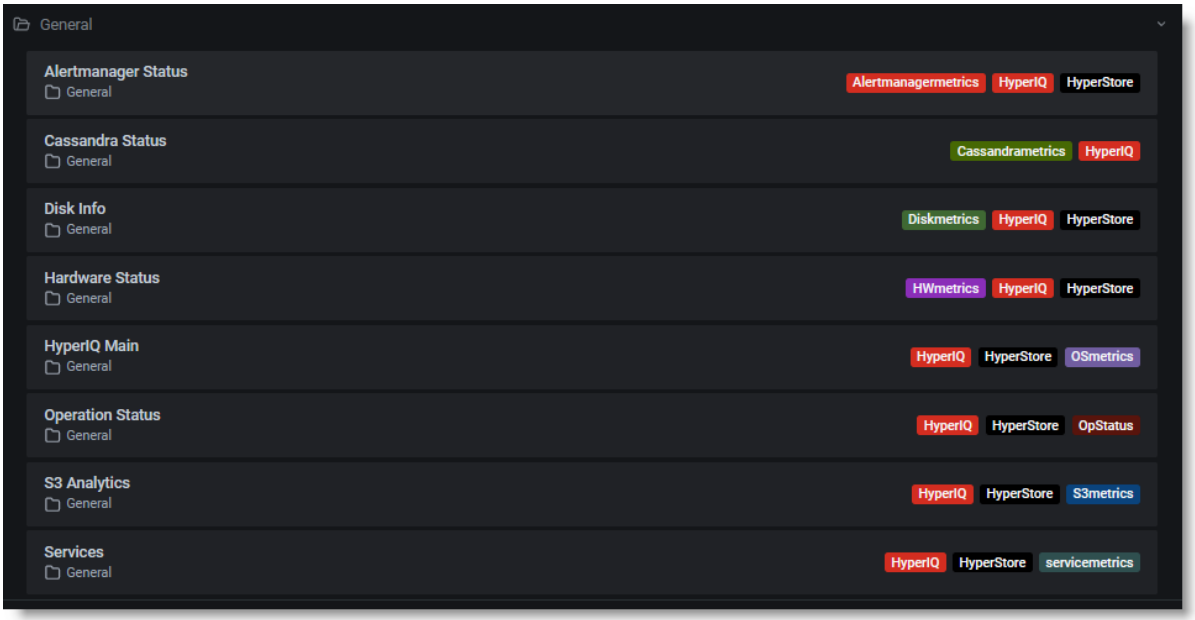
- **HyperIQ Main** -- HyperStore host OS level metrics such as CPU and memory utilization, storage capacity consumption and availability, and front-end and back-end network traffic rates.
- **Hardware Status** -- Intelligent Platform Management Interface (IPMI) sensor data such as hardware component temperatures, fan speeds, and power usage. This data is available in HyperIQ only if your HyperStore host hardware supports IPMI and each HyperStore host has *ipmitool* installed. For example, all HyperStore Appliance machines support IPMI and have *ipmitool* installed.
- **Services** -- HyperStore service level metrics such as up/down status and resource consumption for each service that runs on HyperStore hosts, such as the S3 Service, the Admin Service, Cassandra, and the Redis Credentials DB and Redis QoS DB.
- **Cassandra Status** -- Cassandra database metrics such as those pertaining to space consumption, compaction, and performance. In HyperStore Cassandra is the base technology on which the Metadata DB is built, for storing object metadata and system metadata.
- **Disk Info** -- Usage information for data disks and metadata disks on your HyperStore nodes.
- **Alertmanager Status** -- Summary information about active alerts and notifications.

- **Operation Status** -- Status metrics for HyperStore cluster operations such as repair, cleanup, and rebalance.
- **S3 Analytics** -- HyperStore S3 usage metrics categorized by user, user group, bucket, and S3 operation type (including cross-region replication if applicable). The S3 Analytics dashboard is available only if your HyperStore system license includes support for the Enterprise version of HyperIQ. For information about acquiring a HyperStore license that supports the Enterprise version of HyperIQ, contact Cloudian Support.

To switch between dashboards, in the upper left of the UI click the dashboard title button.

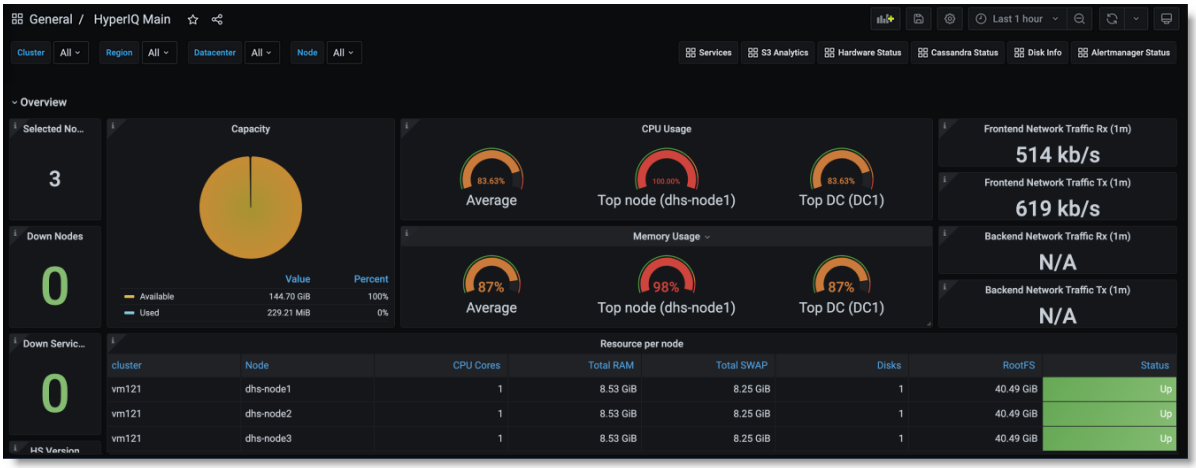


A list of the available dashboards then appears, and you can click on a dashboard name to go to that dashboard.

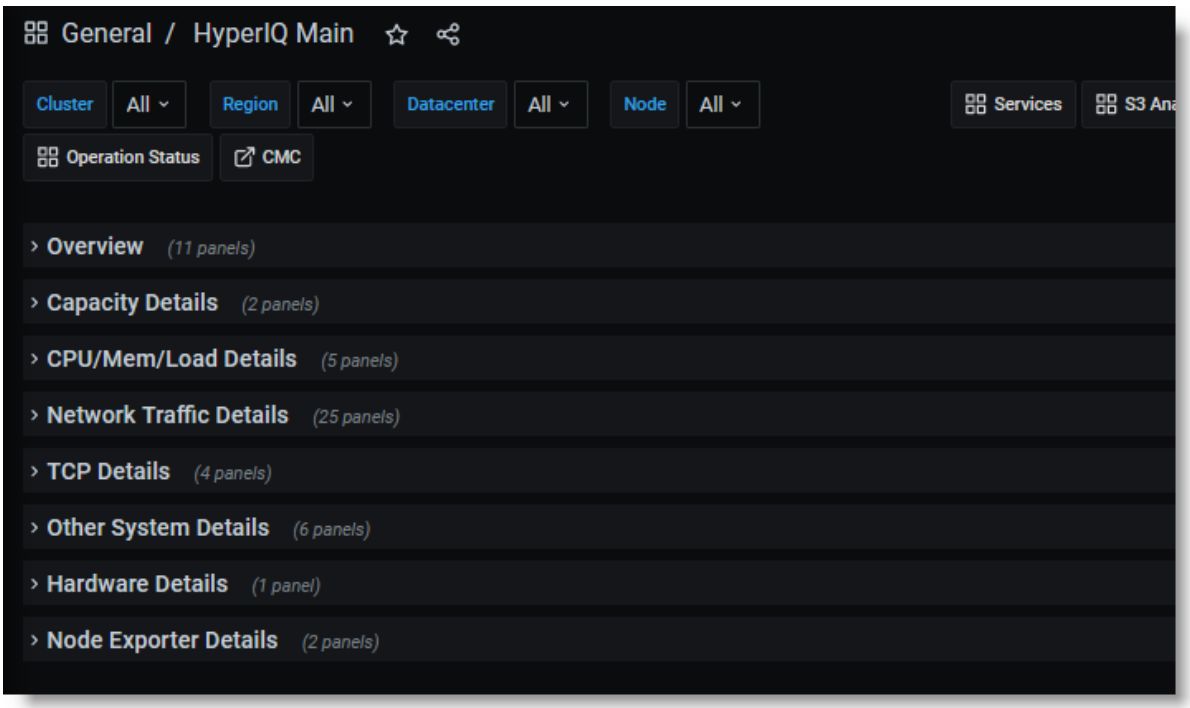


Each dashboard consists of a set of "panels", with each panel providing a graphical representation of one metric (such as CPU Utilization or Top S3 Operations). In each dashboard, panels are grouped into panel folders. When you first access a dashboard, the first panel folder is open so that the panels from that folder display. To see the full list of panel folders available in a dashboard, collapse the open folder.

HyperIQ Main dashboard with **Overview** panel folder open:

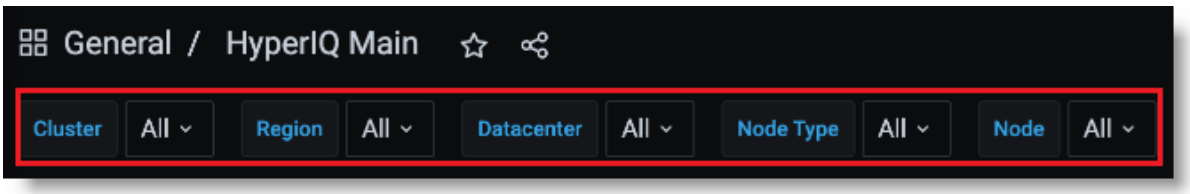


HyperIQ Main dashboard with all panel folders collapsed:

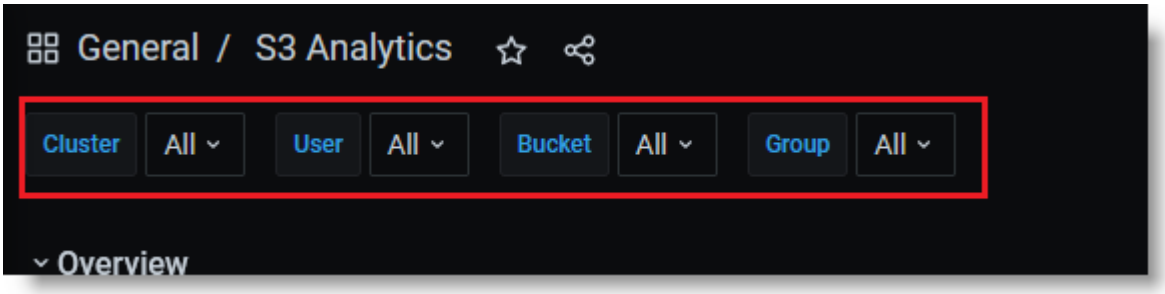


2.2.1. Filtering the Data in a Dashboard

For each dashboard, in the top left of the HyperIQ UI you can filter the data in ways relevant to that dashboard. For example in the **HyperIQ Main** dashboard you can filter data by **cluster**, **service region**, **data center**, **node type**, and **node**. The cluster filter is relevant only if you are using HyperIQ to monitor more than one HyperStore system. The node type filter is relevant only if you are using HyperIQ to monitor HyperStore File nodes and/or HyperStore Search nodes as well as regular HyperStore object storage nodes.



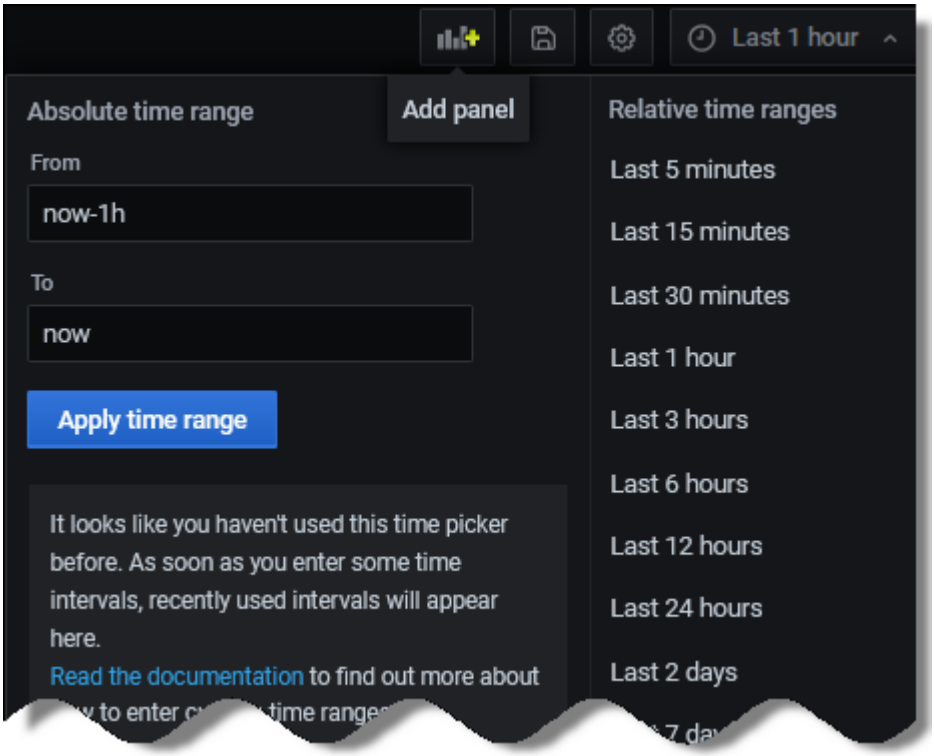
As a second example, in the **S3 Analytics** dashboard you can filter data by **cluster**, **user**, **bucket**, and **group**.



If you filter dashboard data using the filters on the top left of the HyperIQ UI, your chosen filter applies to **all panels in the dashboard**.

2.2.2. Changing the Time Range for a Dashboard

In the top right of the HyperIQ UI, you can change the time period for which a dashboard's panels display historical data. By default, such panels display data for the past hour. If you change the time period the change applies to all time period sensitive panels on the dashboard.



Note By default HyperIQ stores HyperStore monitoring data for only 30 days and so HyperIQ does not support looking back further than 30 days into your HyperStore system's history. The retention period -- and therefore the length of your view back into HyperStore history -- is configurable as described in **"Using the iqmgr Tool to Manage HyperIQ"** (page 27).

2.2.3. Sorting the Data in a Panel

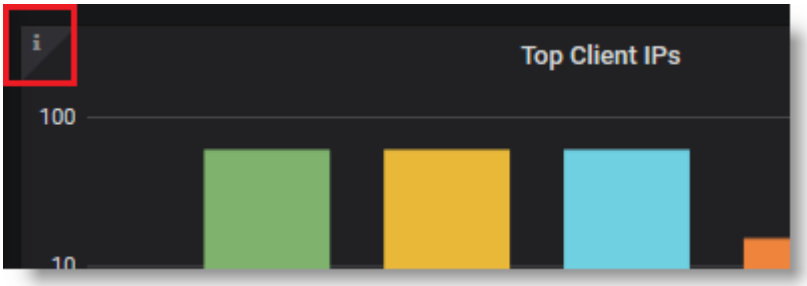
In panels that display data in columns you can use any column as the basis for an alphanumeric sorting of panel data, by clicking on the column heading.

2.2.4. Viewing a Larger Version of a Panel

To enlarge a panel, click the panel title and then click **View**.

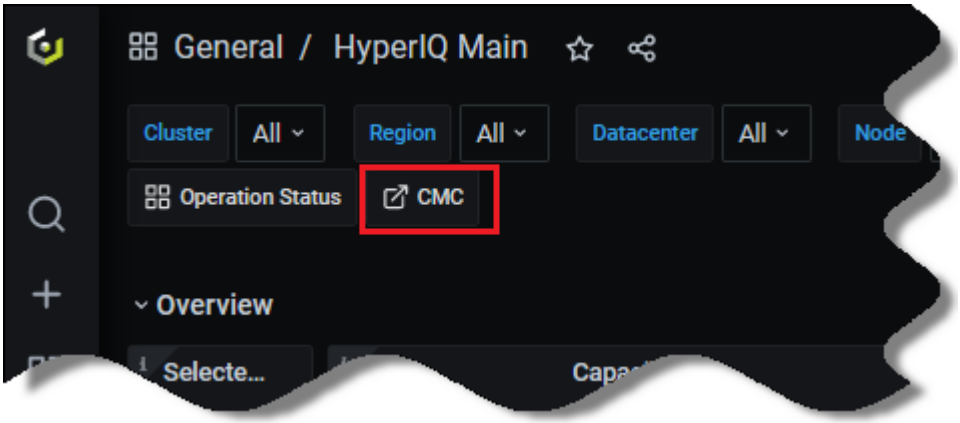
2.2.5. Viewing a Description of the Data in a Panel

For brief information about the data in a panel, hold your cursor over the "i" to the left of the panel name.



2.2.6. Opening the Cloudian Management Console

From each dashboard you can click the **CMC** link to open the Cloudian Management Console for your HyperStore system, in a separate browser tab.



> **Note** For the CMC link to work, your HyperStore system's CMC service endpoint must be configured as an entry in your environment's DNS set-up.

2.2.7. For More Information About Working with Dashboards and Panels

HyperIQ is built on Grafana data visualization technology and exposes much of the Grafana native functionality. For more information about Grafana capabilities see the Grafana official documentation or community resources, which are available on the web.

2.3. Adding HyperIQ Users

The pre-configured user *cloudian_admin* is a "Super Admin" user with full HyperIQ permissions, and is the only user who can add new HyperIQ users. When you are logged in as the *cloudian_admin* user you can create new HyperIQ users and then assign each user one of three roles with differing permission levels:

- **Admin**: Can view, edit, and create dashboards and panels; and manage existing users (such as by changing a user's role or deleting a user). Cannot add new users.
- **Editor**: Can view, edit, and create dashboards and panels. Cannot add or manage users. The pre-configured user *org_editor* has the Editor permission level.
- **Viewer**: Can view dashboards and panels. Cannot edit or create dashboards and panels, or add or manage users.

2.3.1. Adding a HyperIQ User

Note
* By default all new users are created with the **Viewer** role. If you want a user to have more than view permissions, then after creating the user you can explicitly assign them a role as described in **"Assigning a HyperIQ User a Role"** (page 24).
* New users created in the way described below can only log into HyperIQ's Grafana UI (to access dashboards) -- they cannot log into the Alertmanager or Prometheus UIs that are described in the **"Administering HyperIQ"** (page 27) section of this document. **If you want to create a user who can log into the HyperIQ Grafana UI and also the Alertmanager and Prometheus UIs, don't create the user through the HyperIQ Grafana UI** -- instead use the method described in **"Using the iqmgr Tool to Manage HyperIQ"** (page 27).

1. On the left side of the HyperIQ UI, click the Server Admin icon.

2. In the **Server Admin** page, click **New user**.
3. Enter the information for the new user and then click **Create**.

2.3.2. Assigning a HyperIQ User a Role

1. On the left side of the HyperIQ UI, click the Configuration icon.

2. In the **Configuration** page, select the **Users** tab.
3. In the list of users that displays, use the **Role** column to change a user's assigned role.

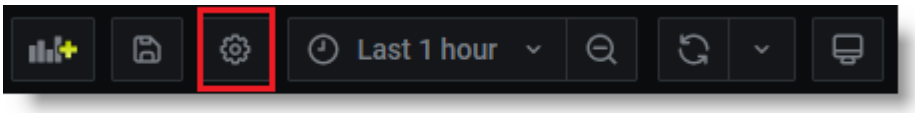
Note The **Configuration** page's **Users** tab also shows an **Invite** button. The **Invite** function -- for sending email invitations that enable recipients to become HyperIQ users if they wish to -- is not supported in HyperIQ.

2.3.3. Assigning Permissions for a Specific Dashboard

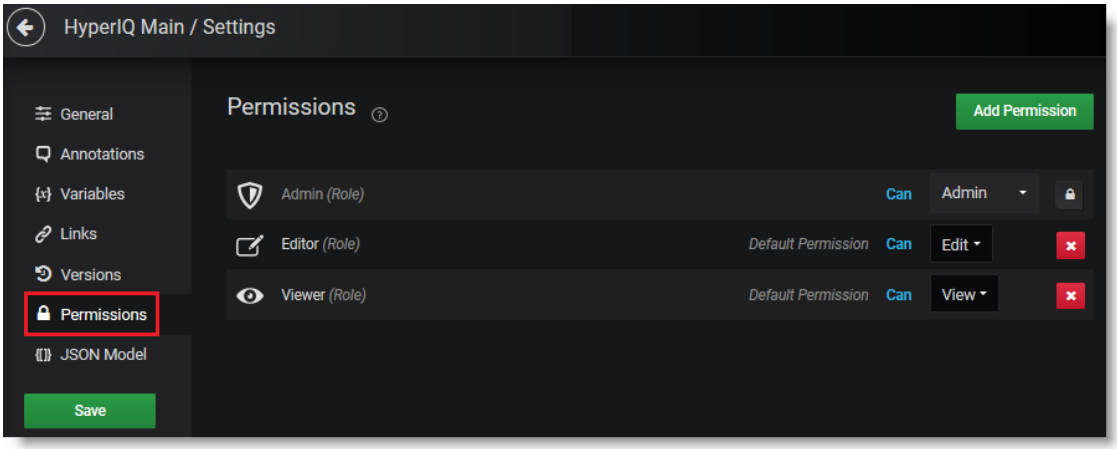
Depending on your HyperIQ use case, simply assigning users one of the three roles (Admin, Editor, or Viewer) may be sufficient for managing HyperIQ user permissions. However, if you wish you can implement more granular permissions, at the dashboard level.

An example would be if you want users with the Viewer role to be able to view only the S3 Analytics dashboard and not the other dashboards. To do so:

1. Go to the HyperIQ Main dashboard, and then in the upper right of the UI click the Dashboard Settings icon:

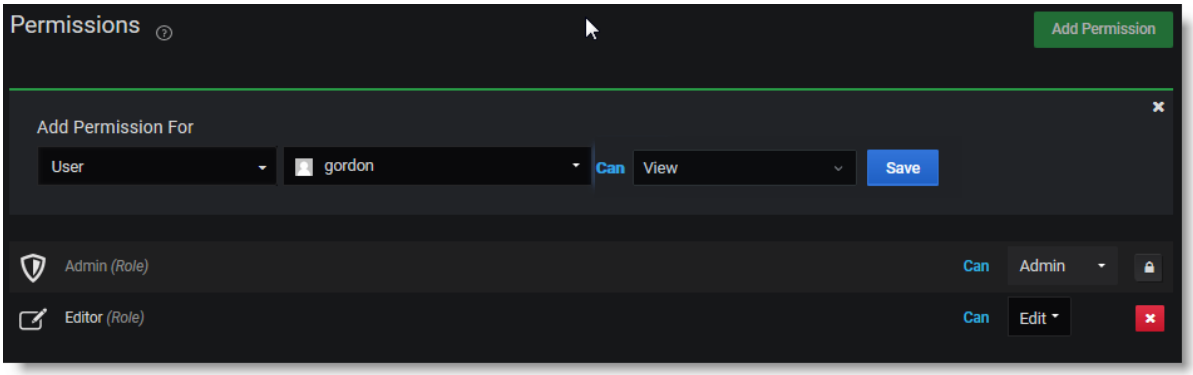


2. In the HyperIQ Main / Settings page select **Permissions**.



3. In the Viewer role row, click the red X at the right to remove the Viewer role from the list of roles that have permissions for this dashboard.
4. On the left side of the HyperIQ Main / Settings page click **Save** (optionally you will be able to enter a comment about the saved change).
5. Repeat Steps 1 through 4 for each of the other dashboards **except** the S3 Analytics dashboard.

As a variation of the example above, you could set things up so that most users with the Viewer role can view only the S3 Analytics dashboard but certain chosen users could also have view-only permissions for the HyperIQ Main dashboard. To do this, follow the steps above, but for the HyperIQ Main dashboard, after removing the Viewer role from the permissions list click **Add Permission**. Then select the user to whom to grant permission, select View as the permission level, and click **Save**.



Note If you plan to make extensive use of dashboard level permissions, you may wish to organize your users into teams, and then after doing so you can set a dashboard level permission for a whole team. To organize users into teams, on the left of the HyperIQ UI click the Configuration icon and then in the **Configuration** page select the **Team** tab.

Note When a dashboard's permissions are configured such that for some users there is a discrepancy between their role-based permission, a team-based permission, and a user-specific permission, the highest degree of permission applicable to the user takes precedence. For example if a user with the Viewer role belongs to a team that has Edit permissions to the dashboard, then the user has Edit permissions for the dashboard.

Chapter 3. Administering HyperIQ

This chapter covers the following topics:

- **"Using the iqmgr Tool to Manage HyperIQ"** (page 27)
- **"Using Docker Commands"** (page 31)
- **"Alerts and Notifications"** (page 32)
- **"Configuring LDAP Authentication for HyperIQ Users"** (page 47)
- **"Uninstalling HyperIQ"** (page 49)

3.1. Using the *iqmgr* Tool to Manage HyperIQ

HyperIQ includes a management tool named *iqmgr*. The *iqmgr* tool is stored under this directory on your HyperIQ node:

```
$HOME/.config/cloudian/hyperiq
```

- If you installed HyperIQ as a virtual appliance, you can run *iqmgr* from any directory.

  > **Note** To log into the HyperIQ virtual appliance, use the username **cloudian** and the virtual appliance login password that you set when you installed the virtual appliance.

- If you installed HyperIQ in an existing Docker environment, you must change into the *$HOME/.-config/cloudian/hyperiq* directory to run *iqmgr*, or else include that directory path when running *iqmgr*. (Or for convenience you could, for example, create a symlink to the tool.)

With the *iqmgr* tool you can perform several types of HyperIQ management tasks:

- **"Updating HyperIQ to Monitor Additional HyperStore Resources"** (page 27)
- **"Changing HyperIQ Configuration Settings"** (page 28)
- **"Managing HyperIQ/Alertmanager/Prometheus Users"** (page 30)
- **"Additional iqmgr Commands"** (page 30)

For some simple *iqmgr* commands you will specify command arguments, and simply execute the command. For *iqmgr* command that perform more complex tasks, when you launch the command you will be led through a series of interactive prompts.

3.1.1. Updating HyperIQ to Monitor Additional HyperStore Resources

Update HyperIQ because you've made changes to the HyperStore cluster that HyperIQ has been monitoring:

```
iqmgr setup [-k KEY_FILENAME] [--auxiliary-nodes] [--with-managed-services] [--secondary-hiq]
```

Do this if you've changed your HyperStore cluster composition. For example, run *iqmgr setup* if you:

- Add a node or nodes to your HyperStore cluster
- Remove a node from your HyperStore cluster

- Add HyperStore File nodes and/or HyperStore Search nodes to your HyperStore cluster (and you want to monitor these "auxiliary" nodes)
- Add a HyperBalance load balancer to your HyperStore cluster (and you want to monitor HyperBalance)

> **Note** Before updating your HyperIQ setup to monitor HyperBalance you must enable the Prometheus exporter on each of your HyperBalance nodes. On each HyperBalance node, log into the HyperBalance UI, go to **Cluster Configuration -> Layer 7 -- Advanced Configuration** and then check the "Enable Prometheus Exporter" checkbox.

For descriptions of the command options see **"Integrating the HyperIQ VM with HyperStore"** (page 11).

> **Note** When prompted for whether you want to refresh the local *cloudianservicemap.json* file, answer *yes* (unless you have HyperStore on a private network to which HyperIQ is not attached and you've manually edited HyperIQ's copy of the service map file as described in the HyperIQ installation procedure).

Have HyperIQ monitor an additional HyperStore cluster:

```
iqmgr config cluster add [-k KEY_FILENAME] [--auxiliary-nodes] [--with-managed-services]
[--secondary-hiq]
```

A "cluster" here means an entire HyperStore system, which may encompass one or more data centers and one or more service regions. So you would use this option if you want to have HyperIQ monitor an additional entire HyperStore system, as well as the HyperStore system that it's already been monitoring.

For descriptions of the command options see **"Integrating the HyperIQ VM with HyperStore"** (page 11).

Remove HyperIQ monitoring from a HyperStore cluster:

```
iqmgr config cluster remove
```

3.1.2. Changing HyperIQ Configuration Settings

Change the *cloudian_admin* user's password for logging in to the HyperIQ Grafana UI:

```
iqmgr config password
```

When you run this command you will be prompted to supply the new password. Do not include the password on the command line when launching the command.

This password is also used for logging in to the Alertmanager and Prometheus UIs.

> **Note** Use this method for changing the *cloudian_admin* user's password -- **not** the "Change Password" method that is available in the lower left of the HyperIQ UI. For other users (including the pre-configured *org_editor* user), the Change Password method in the UI is the correct method to use.

Configure S3 analytics alerts, or configure alert notifications, or configure LDAP integration for the admin user:

```
iqmgr config {set,unset,apply,get,list}
```

For detail see **"Configuring Alerts and Notifications"** (page 37) and **"Configuring LDAP Authentication for HyperIQ Users"** (page 47).

Configure for how many days to retain HyperStore monitoring data in the HyperIQ database:

```
iqmgr config retention <n>d
```

Your <*n*> value must be an integer. For example, to retain day for 60 days:

```
iqmgr config retention 60d
```

After the retention period the data is automatically deleted. The default is 30 days.

> **Note** The data retention period will determine how far back into time you can look when analyzing HyperStore statistics through HyperIQ. The longer the retention period, the more storage capacity is required for the HyperIQ database. Consult with Cloudian Support if you have questions about HyperIQ storage sizing requirements for your use case and desired data retention period.

Change the HyperIQ UI listening port numbers:

```
iqmgr httpcfg --http-port <integer>
```

```
iqmgr httpcfg --https-port <integer>
```

By default the HTTP port is 3000 and the HTTPS port is 3443.

If you change the listening port numbers for the HyperIQ main UI the change will apply also to the Alert-manager and Prometheus UIs.

> **IMPORTANT !** The HyperIQ HTTPS port is also used by HyperStore nodes, for transmitting data from HyperStore to HyperIQ. Make sure that your network allows each HyperStore node to access HyperIQ's HTTPS port.

Configure HyperIQ to listen only on HTTPS and not on regular HTTP:

```
iqmgr httpcfg -s
```

> **Note** HyperIQ's HTTPS listener will accept connections from browsers using TLS versions 1.2 or 1.3. Connections using TLS 1.0 or 1.1 are not allowed.

Create a New Self-Signed TLS Certificate for the HyperIQ HTTPS listener:

```
iqmgr httpcfg --self-signed
```

HyperIQ automatically generates a self-signed TLS certificate to be used by the HyperIQ HTTPS listener. You can run the command above to trigger HyperIQ to generate a new self-signed certificate and use the new certificate for its HTTPS listen.

Have the HyperIQ HTTPS listener use a TLS certificate that you provide:

```
iqmgr httpcfg -c <full path to cert file> -k <full path to key file>
```

You must copy the certificate file and private key file on to the HyperIQ node before running the command above. The certificate and private key must be in .PEM format and the key must not have a passphrase.

Change HyperIQ virtual appliance network settings:

```
sudo iqmgr network config
```

This command for changing network settings is applicable only to the virtual appliance and not to HyperIQ installed in an existing Docker environment.

To check the current network settings for the virtual appliance:

```
iqmgr network status
```

3.1.3. Managing HyperIQ/Alertmanager/Prometheus Users

Add a user, delete a user, or list users:

```
iqmgr user add <username>
```

```
iqmgr user delete <username>
```

```
iqmgr user list
```

This way of adding users creates users who can access the HyperIQ Grafana UI **and also** the Alertmanager and Prometheus UIs (described in **"Alerts and Notifications"** (page 32)). If you want to create users who can access only the HyperIQ Grafana UI, don't use *iqmgr* -- instead use the Grafana UI's user management page to create the users as described in **"Adding a HyperIQ User"** (page 24).

Note that:

- When you run any of these commands you will be prompted to provide the *cloudian_admin* user's password. To avoid being prompted for this password, you can include the argument *-g <password>* when running the command, where *<password>* is the *cloudian_admin* user's password.
- When you run the *iqmgr user add* command you will be prompted to create (enter) a password for the new user. To avoid being prompted for this password, you can include the argument *-p <password>* when running the command, where *<password>* is the password to assign to the new user.
- Users that you create with the *iqmgr user add* command will have the Viewer role in the HyperIQ Grafana UI. (After creating users you can optionally revise their role as described in **"Assigning a HyperIQ User a Role"** (page 24).)
- The *iqmgr user list* command returns a list of "Grafana users" and a list of "Nginx users" (Alertmanager and Prometheus UI users). Users that you create through the *iqmgr user add* command will appear in both lists. Users that you create through the Grafana UI will appear only in the "Grafana users" list.
- If you want to delete a user who you had created through the *iqmgr user add* command, do it with the *iqmgr user delete* command so that the user is deleted as a Grafana user and also as an Alertmanager and Prometheus UI user. If you instead delete such a user through the Grafana UI's user management page, the user will only be deleted as a Grafana user and will persist as an Alertmanager and Prometheus UI user.

Change the Alertmanager and Prometheus UI password for a user other than the *cloudian_admin* user.

```
iqmgr user password <username>
```

Note that:

- When you run *iqmgr user password <username>* you will be prompted to enter the new password. Do not include the password on the command line when launching the command.
- This command does not change the user's Grafana password. To change the Grafana password for a user other than the *cloudian_admin* user, use the user management page in the Grafana UI.
- To change the *cloudian_admin* user's password for both the Grafana UI and the Alertmanager/Prometheus UIs, use *iqmgr config password* as described earlier in this table.

3.1.4. Additional iqmgr Commands

Gather and package HyperIQ container logs:

```
iqmgr logs
```

This gathers all HyperIQ related log files into a tarball named *hyperiq-logs-<timestamp>.tgz* under the current directory (the directory in which you run *iqmgr logs*). You may need to do this if you are working with Cloudian Support to troubleshoot a HyperIQ issue.

Copy a file to a specified container:

```
iqmgr cp <source-filepath> <container>:<destination-path>
```

Note that:

- For this command and for the two container-related commands that follow below, you do not need to include "hyperiq_" in the *<container>* name. For example, identify the Prometheus container as *prometheus*, not as *hyperiq_prometheus*.
- If you have a HyperStore Single-Node system and you are logged into the HyperStore/HyperIQ node as the HyperStore Shell user *sa_admin*, copying is only allowed between a container and the */home/sa_admin* directory (or sub-directories under that directory)

Copy a file from a specified container:

```
iqmgr cp <container>:<source-filepath> <destination-path>
```

Run a command inside a specified container:

```
iqmgr exec <container> <command>
```

Display the HyperIQ version number:

```
iqmgr version
```

Display *iqmgr* command help:

```
iqmgr <command> --help
```

3.2. Using Docker Commands

HyperIQ is implemented as a containerized application that uses Docker for containerization and Docker Compose for container orchestration. This is the case even if you installed HyperIQ as a virtual appliance (Docker and Compose are packaged with the appliance).

HyperIQ consists of several containers including:

- *hyperiq_grafana* (Grafana UI)
- *hyperiq_prometheus* (Prometheus database)
- *hyperiq_alertmanager* (Prometheus Alertmanager, for alert handling)
- *hyperiq_nginx* (web server used for Prometheus and Alertmanager UIs)
- *hyperiq_iqmgr* (HyperIQ management service)

You can check on and manage the HyperIQ containers using standard *docker* and *docker-compose* commands. You can run such commands from the operating system command prompt of the HyperIQ host machine. For information about supported commands enter *docker --help* (or for *docker-compose* commands, enter *docker-compose --help*). You can also refer to Docker's online documentation for more detailed information and context.

For example, to list the currently running HyperIQ containers and their health status you can use the command ***docker ps***.

```
cloudian@hyperiq:~$ docker ps
CONTAINER ID     IMAGE       COMMAND
         CREATED        STATUS                  PORTS        NAMES
bfeff9d8f60d     quay.io/cloudian/hyperiq-iqmgr:1.6.0-alpha      "/bin/sh -c 'sleep 1…"
         3 hours ago    Up 3 hours (healthy)                 hyperiq_iqmgr
43d9d12dafb3     quay.io/cloudian/hyperiq-alertmanager:1.6.0-alpha   "/bin/alertmanager -…"
         3 hours ago    Up 3 hours (healthy)    9093/tcp     hyperiq_alertmanager
abbfa2357d5d     quay.io/cloudian/hyperiq-prometheus:1.6.0-alpha     "/bin/prometheus --c…"
         3 hours ago    Up 3 hours (healthy)    9090/tcp     hyperiq_prometheus
9b4e64b5a4ef     quay.io/cloudian/hyperiq-grafana:1.6.0-alpha        "/run.sh"
         3 hours ago    Up 3 hours (healthy)    3000/tcp     hyperiq_grafana
...
...
$
```

Another useful command is ***docker restart***, to restart a particular container. For example, to restart the Grafana UI container:

```
cloudian@hyperiq:~$ docker restart hyperiq_grafana
hyperiq_grafana
cloudian@hyperiq:~$
```

You should rarely need to access the interior of one of the HyperIQ containers, but it may be useful on occasion -- such as if you want to edit the main Grafana configuration file *grafana.ini*. Here the ***docker exec*** command is used to enter the Grafana container as *root*, then the *grafana.ini* file is edited using *vi*, then the container is exited, and then the container is restarted to apply the configuration change:

```
cloudian@hyperiq:~$ docker exec -u root -it hyperiq_grafana sh
/usr/share/grafana # vi /etc/grafana/grafana.ini
/usr/share/grafana # exit
cloudian@hyperiq:~$ docker restart hyperiq_grafana
hyperiq_grafana
cloudian@hyperiq:~$
```

HyperIQ also supports using ***docker-compose*** commands. To successfully run *docker-compose* commands you must be in the HyperIQ configuration directory (*$HOME/.config/cloudian/hyperiq* on Linux). For a list of supported commands enter *docker-compose --help*.

3.3. Alerts and Notifications

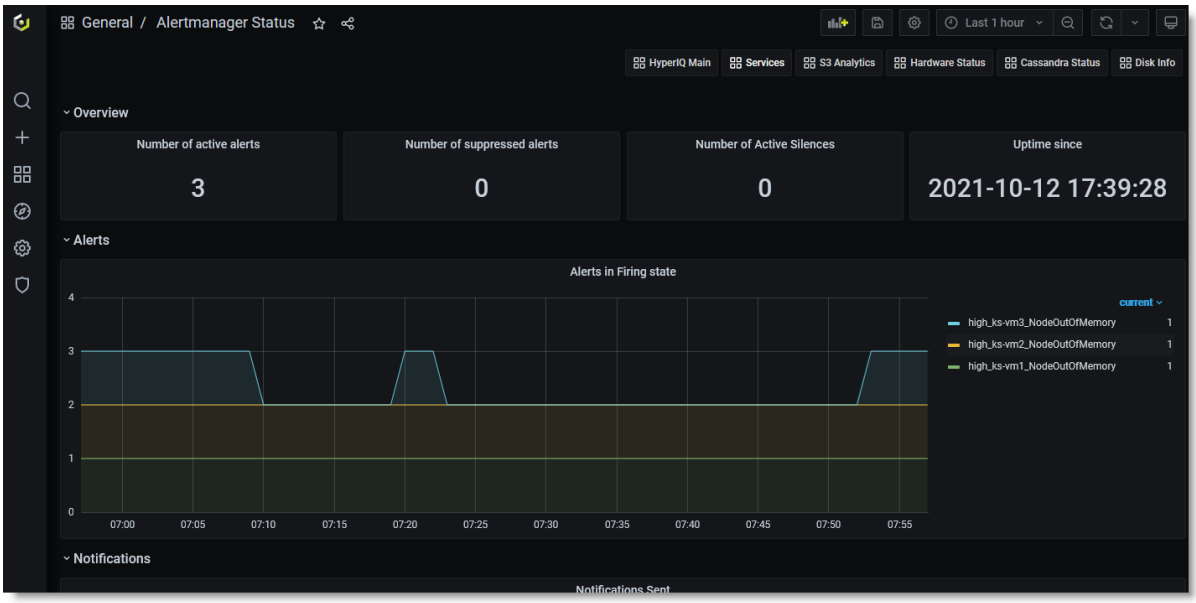
This section covers the following topics:

- **"Viewing Actively Firing Alerts"** (page 32)
- **"Viewing the Full List of Enabled Alerts"** (page 34)
- **"Silencing Alerts"** (page 34)
- **"Configuring Alerts and Notifications"** (page 37)
- **"Alert Codes"** (page 45)

3.3.1. Viewing Actively Firing Alerts

You can view actively firing alerts in HyperIQ's **Alertmanager Status** dashboard. Here you can see the currently active alerts as well as the current number of alerts that are "suppressed". Suppressed alerts are alerts that either are currently being inhibited by pre-configured inhibition rules (such as not generating service down

alerts for a node for which there is currently a node down alert) or are currently being "silenced" by a silence that you have set in the Alertmanager UI (see **"Silencing Alerts"** (page 34)). You can also see how many alert notifications have been sent (applicable only if you have enabled notifications as described in **"Administering HyperIQ"** (page 27)).



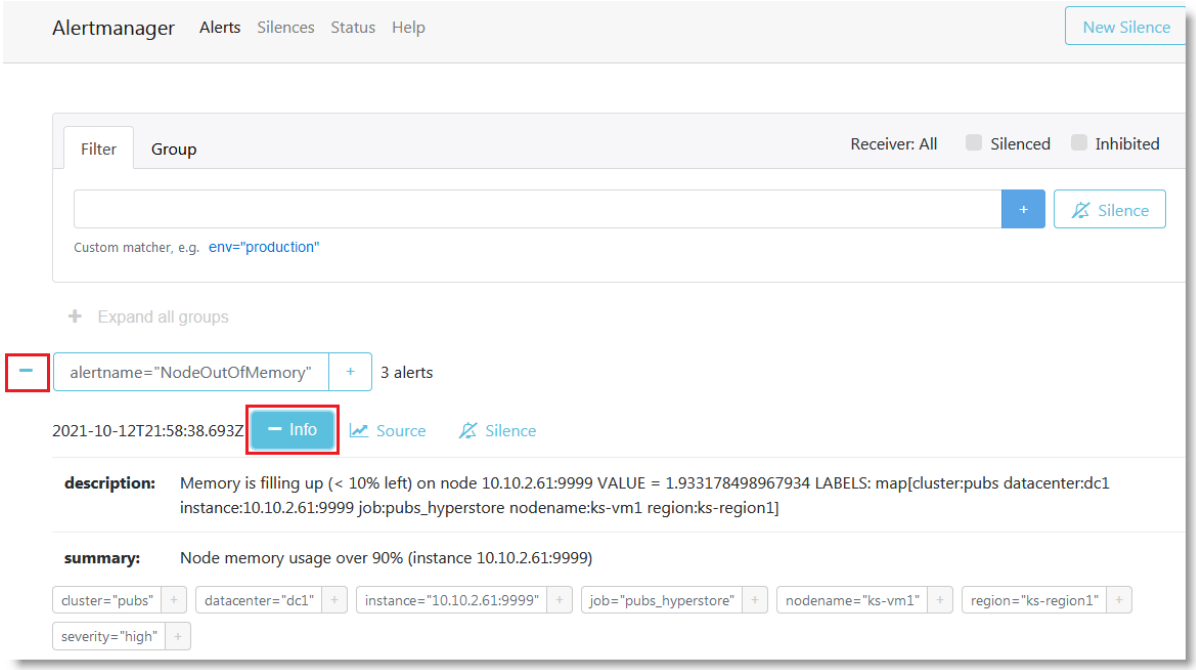
To see more detail about a particular alert, access the Alertmanager UI at either of these URIs:

http://<HyperIQ IP address>:3000/alertmanager

https://<HyperIQ IP address>:3443/alertmanager

You will be prompted for HTTP Basic Authentication credentials: enter username **cloudian_admin** and the HyperIQ UI password that you set during the HyperIQ set-up process.

In this Alertmanager UI example we have drilled into **Info** about one of the active *NodeOutOfMemory* alerts:



For some alerts (such as the one shown above), the threshold that triggers the alert is stated in the **Info** that you can see in the Alertmanager UI. If the triggering threshold is not shown there for a particular alert, you can view that information in the Prometheus UI at either of these URIs:

http://<HyperIQ IP address>:3000/prometheus

https://<HyperIQ IP address>:3443/prometheus

Again the HTTP Basic Authentication credentials are username **cloudian_admin** and your HyperIQ UI password.

Select the **Alerts** tab and then select the alert that you are interested in. In this Prometheus UI example we can see that the *DataDiskOutOfSpace* alert is triggered when remaining space on a particular data disk device falls below 20%. We can also see in this view that the alert code for this alert is GS000004.

3.3.2. Viewing the Full List of Enabled Alerts

To view a list of all alerts that are enabled in HyperIQ -- not just alerts that are currently firing but all alerts for which triggering criteria have been defined and are being monitored by the system -- access the Prometheus UI:

http://<HyperIQ IP address>:3000/prometheus

https://<HyperIQ IP address>:3443/prometheus

Log in with username **cloudian_admin** and your HyperIQ UI password, then select the **Alerts** tab. This displays the full list of enabled alerts, and you can click on an alert for details about how the alert is defined.

Note that each alert has a unique **alert code**, and an alert's code is among the alert details you can see in the Prometheus UI. For alerts that may appear both in HyperIQ and in HyperStore's Cloudian Management Console (CMC), the same alert code is used by both applications. For example, in the CMC the alert code when the S3 Service is down on a node is SS000007, and in HyperIQ the alert code when the S3 Service is down on a node is also SS000007. For a complete list of HyperIQ alert codes see **"Alert Codes"** (page 45).

> **Note** S3 Analytics alerts are disabled by default. For information about how to enable S3 Analytics alerts see **"Configuring Alerts and Notifications"** (page 37).

3.3.3. Silencing Alerts

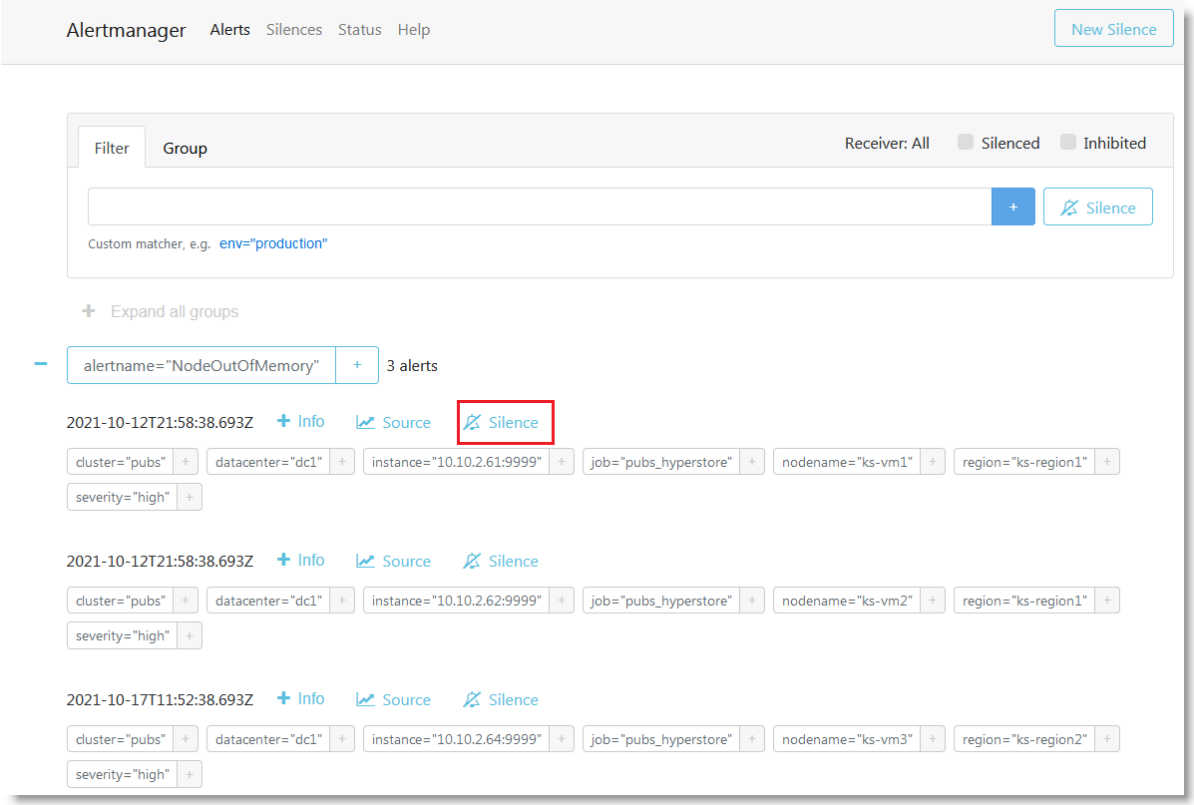
"Silencing" is a way of muting an alert so that no notifications are sent out for that alert for a time duration of your choosing. To silence an alert, access the Alertmanager UI at either of these URIs:

http://<HyperIQ IP address>:3000/alertmanager

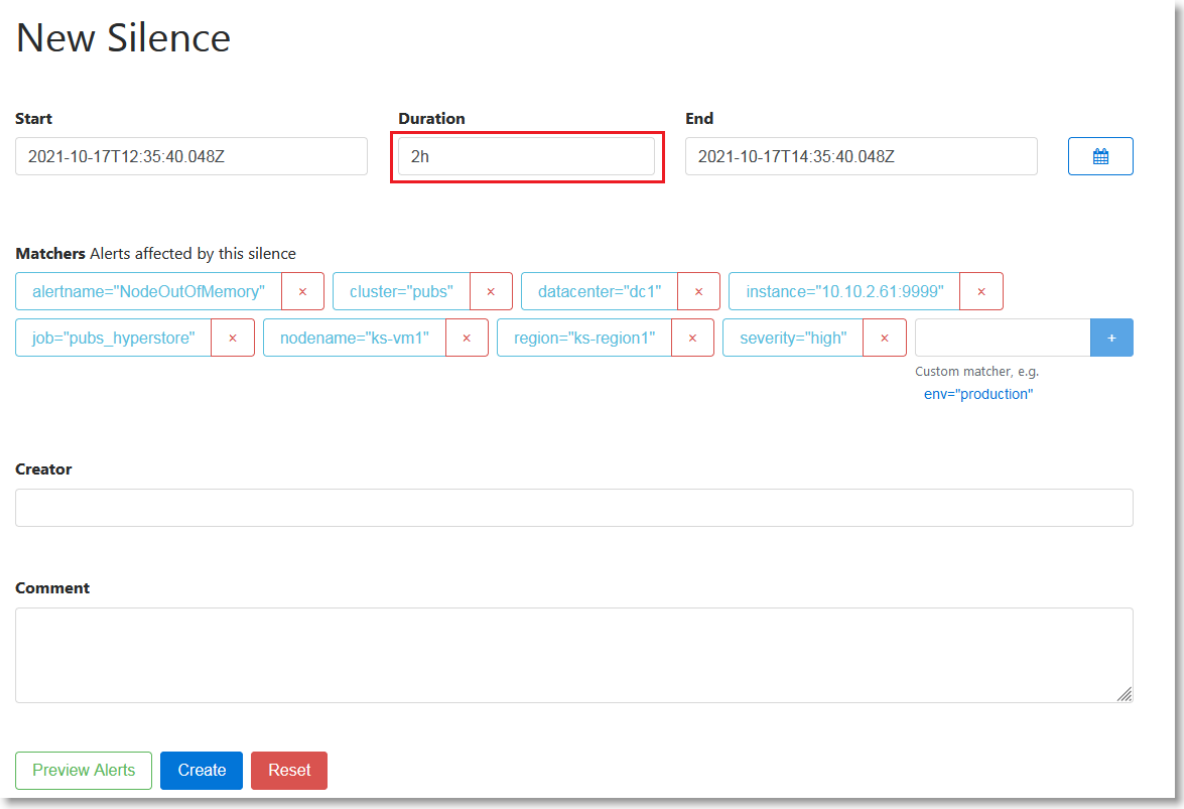
https://<HyperIQ IP address>:3443/alertmanager

You will be prompted for HTTP Basic Authentication credentials: enter username **cloudian_admin** and the HyperIQ UI password that you set during the HyperIQ set-up process.

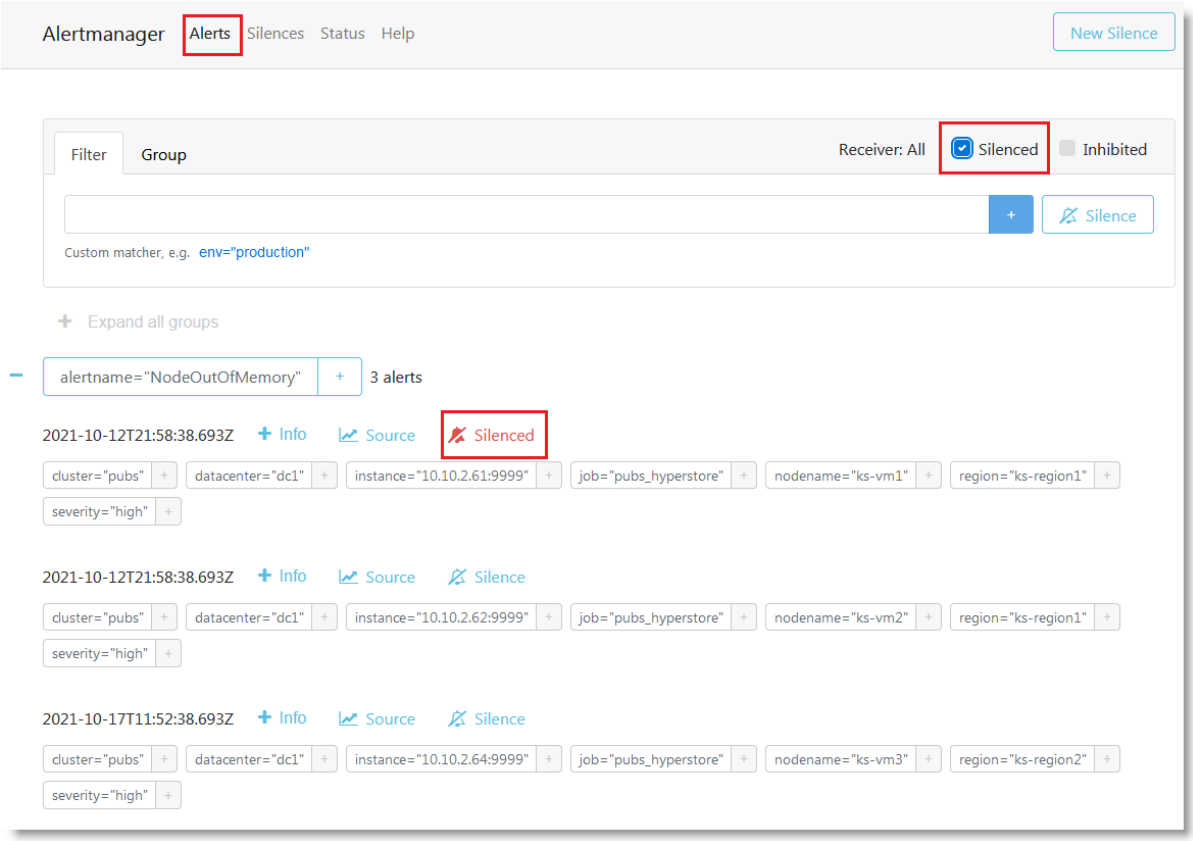
In the Alertmanager UI, select the alert that you want to silence and then click **Silence**.



In the **New Silence** screen, set your desired duration for the silence (the default is "2h", meaning two hours). If you want to silence the alert indefinitely you can set the duration to a very high value (for example, "10000d"). Also, if you wish you can expand the scope of the silence by deleting one or more of the match criteria that define the silence, by clicking the small red X beside that match criteria. For example if you delete the *cluster* and *datacenter* and *instance* (IP address) and *nodename* criteria the silence will apply to **all** clusters and **all** datacenters and **all** nodes. Complete the "Creator" and "Comment" fields, and then click **Create** to create the silence.



After configuring the silence you can return to the **Alerts** tab, select the filter to show **Silenced** alerts, and see the alerts that are now silenced.



To end a Silence you can click the **Silenced** button, and then in the screen that displays click **Expire**.

> **Note** To proactively create a Silence for an alert that is not currently firing, in the upper right of the Alertmanager UI click **New Silence**. In the **Matchers** list, enter *alertname="<Name of Alert>"* (for example, *alertname="NodeOutOfMemory"*) and whatever other matchers you want to use to define the silence (for example *datacenter="DC1"*).

3.3.4. Configuring Alerts and Notifications

Using the *iqmgr config* command you can configure HyperIQ alerting and notification behavior. The *iqmgr config* command syntax is:

Command	Description
iqmgr config set <setting> <value> **Note** For settings that are for changing a password do not include the password value on the command line. Instead run *iqmgr config set <setting>* with no value, and you will be prompted to provide the new password.	Set the specified setting to the specified value. After running *iqmgr config set* you must run *iqmgr config apply* to apply the setting change to the system. The supported settings are described in these sections that follow further below: ◦ **"S3 Analytics Alert Threshold Settings"** (page 38) ◦ **"Node and Disk Alert Threshold Settings"** (page 39) ◦ **"NTP Drift Alert Threshold Setting"** (page 40) ◦ **"General Notification Settings"** (page 40) ◦ **"Email Notification Settings"** (page 42) ◦ **"Slack Notification Settings"** (page 45) ◦ **"MS Teams Notification Setting"** (page 45) ◦ **"Splunk Notification Settings"** (page 45)
iqmgr config unset <setting>	Revert the specified setting to its default condition, which in most cases is to have no value (i.e., the setting is disabled by default). After running *iqmgr config unset* you must run *iqmgr config apply* to apply the setting change to the system.
iqmgr config apply	Apply a configuration change that you have made by the *iqmgr config set* command or the *iqmgr config unset* command. If you are making multiple setting changes you can run a *iqmgr config set* command for each change and then run *iqmgr config apply* once afterwards to apply all the changes. If you do not run the *iqmgr config apply* command your setting change (s) will not take effect in the system.
iqmgr config get [<setting>|<setting prefix>|ALL]	Get the current value of a single specified setting, or of a group of settings identified by prefix, or of all S3 alerts settings and notification settings. The supported prefixes for retrieving a group of settings are:

Command	Description
	◦ */alerts/s3* -- Returns all S3 alerts settings ◦ */alerts/node* -- Returns all node alerts settings ◦ */alerts/disk* -- Returns all disk alerts settings ◦ */alertmanager/notifications/email* -- Returns all email notification settings ◦ */alertmanager/notifications/slack* -- Returns all Slack notification settings ◦ */alertmanager/notifications* -- Returns all notification settings
iqmgr config list	Get the current value of all S3 alerts settings and notifications settings along with very brief descriptions of each setting.
iqmgr config --help	Get inline help for iqmgr config options.

Note For background information on how to launch the *iqmgr* tool in your particular operating system, see **"Using the iqmgr Tool to Manage HyperIQ"** (page 27).

3.3.4.1. S3 Analytics Alert Threshold Settings

By default, S3 Analytics alerts -- alerts based on your users' utilization of HyperStore's S3 Service -- are disabled. This is because there are no universally appropriate thresholds for such alerts, and instead such thresholds should be based on your system's expected S3 workload and your preferences regarding what levels of service utilization merit alerts. You can enable S3 alerts by using the *iqmgr config set* command to assign threshold values to some or all of the settings listed below.

Note Enabling S3 Analytics based alerts is supported **only if your HyperStore license supports HyperIQ Enterprise**.

The table below lists the supported S3 alert settings. For each alert that you want to enable, specify the threshold *<value>* as a simple integer. For alerts based on a data size threshold, specify the value as a number of bytes -- for example, *100000* not *100KB*.

S3 Alert Threshold Setting	Description
/alerts/s3/503s_last_10_min_limit	If your HyperStore system as a whole returns more than this many HTTP 503 responses to S3 clients in a 10 minute interval, the alert is triggered.
/alerts/s3/delete_transactions_per_hour	If your HyperStore system as a whole processes more than this many S3 DELETE transactions in a one hour interval, the alert is triggered.
/alerts/s3/get_throughput_limit	If the S3 GET throughput for your HyperStore system as a whole exceeds this many bytes per second as cal-

S3 Alert Threshold Setting	Description
	culated over a 15 minute interval, the alert is triggered.
/alerts/s3/get_transactions_per_second	If your HyperStore system as a whole processes more than this many S3 GET transactions per second as calculated over a 15 minute interval, the alert is triggered.
/alerts/s3/put_throughput_limit	If the S3 PUT throughput for your HyperStore system as a whole exceeds this many bytes per second as calculated over a 15 minute interval, the alert is triggered.
/alerts/s3/put_transactions_per_second	If your HyperStore system as a whole processes more than this many S3 PUT transactions per second as calculated over a 15 minute interval, the alert is triggered.
/alerts/s3/total_system_bytecount_limit	If the net storage bytes count in your HyperStore system as a whole exceeds this number, the alert is triggered. This alert is based on **net** bytes of object data stored in the system, which excludes overhead from replication or erasure coding.
/alerts/s3/total_system_objectcount_limit	If the object count in your HyperStore system as a whole exceeds this number, the alert is triggered.
/alerts/s3/user_30_day_download_limit	If any single user downloads more than this many bytes of object data in a 30 day period, the alert is triggered.
/alerts/s3/user_30_day_upload_limit	If any single user uploads more than this many bytes of object data in a 30 day period, the alert is triggered.

For example:

```
iqmgr config set /alerts/s3/user_30_day_download_limit 100000000000
iqmgr config set /alerts/s3/user_30_day_upload_limit 100000000000
iqmgr config apply
```

After enabling S3 alerts, access the Prometheus UI at either of these URIs:

http://<HyperIQ IP address>:3000/prometheus

https://<HyperIQ IP address>:3443/prometheus

The HTTP Basic Authentication credentials are username **cloudian_admin** and your HyperIQ UI password.

In the Prometheus UI select the **Alerts** tab. In the list of enabled alerts you should see the S3 alerts that you enabled.

3.3.4.2. Node and Disk Alert Threshold Settings

Node and disk alerts are enabled by default but if you wish you can customize the thresholds for any of these alerts. These alerts are all per-node and per-device (for example, per each HyperStore data disk on each node).

Node or Disk Alert Threshold Setting	Description
/alerts/node/oom_threshold	NodeOutOfMemory alert threshold (% used; default 90)
/alerts/node/high_cpu_threshold	HighCPUUsage alert threshold (% used; default 90)

Node or Disk Alert Threshold Setting	Description
/alerts/node/high_inbound_traffic_threshold	NodeHighInboundNetworkTraffic alert threshold (% NIC receiving capacity used, default 70)
/alerts/node/high_outbound_traffic_threshold	NodeHighOutboundNetworkTraffic alert threshold (% NIC transmitting capacity used, default 70)
/alerts/disk/data_disk_outof_space_threshold	DataDiskOutOfSpace alert threshold (% space remaining, default 20)
/alerts/disk/root_disk_outof_space_threshold	RootDiskOutOfSpace alert threshold (% space remaining, default 10)

For example:

```
iqmgr config set /alerts/disk/data_disk_outof_space_threshold 25
iqmgr config apply
```

3.3.4.3. NTP Drift Alert Threshold Setting

By default an alert will be triggered if a HyperStore node's system time differs from the HyperIQ node's system time by more than 5 seconds. If you wish you can change the time difference that triggers the alert, using the setting below.

NTP Alert Threshold Setting	Description
/alerts/node/node_time_out_of_sync_threshold	NodeTimeOutOfSyncWithHIQ alert threshold (seconds)

For example:

```
iqmgr config set /alerts/node/node_time_out_of_sync_threshold 3
iqmgr config apply
```

Note An alert will also be triggered if the *ntpd* service does down on a HyperStore node.

3.3.4.4. General Notification Settings

To configure general notification settings -- which apply to all the notification channels that you set up, such as email or Slack -- use *iqmgr config set* commands to assign values to some or all of the settings listed below. Remember to subsequently run *iqmgr config apply*.

General Notification Setting	Description
/alertmanager/notifications/send_resolved	Optional, defaults to false. If set to *true*, HyperIQ will send a notification when an alert condition is resolved (for example, when a service that has been down comes back up again).
/alertmanager/custom/external_url	Optional, defaults to *http://<HyperIQ_IPad-dress>:3000*. Alert notifications include a link back to the Alertmanager UI, at this URL.

General Notification Setting	Description
	You can use the */alert-manager/custom/external_url* setting if you want to specify an Alertmanager URL that's different than the default -- for example, if you've configured a HyperIQ entry in your DNS system and you want to specify that hostname in place of the HyperIQ IP address segment of the URL.
/alertmanager/notifications/urgent_group_wait	Optional, defaults to 10s. How long to initially wait to send a notification for a group of similar urgent alerts. Allows time to wait for an inhibiting alert to arrive, or to collect more initial alerts for the same group before sending the notification. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/urgent_group_interval	Optional, defaults to 30s. How long to wait before sending a notification about new urgent alerts that are added to a group of urgent alerts for which an initial notification has already been sent. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/urgent_repeat_interval	Optional, defaults to 30m. How long to wait before sending a notification again if it has already been sent successfully for an urgent alert. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/high_group_wait	Same as *urgent_group_wait*, except for high severity alerts rather than urgent severity alerts. Defaults to 5m.
/alertmanager/notifications/high_group_interval	Same as *urgent_group_interval*, except for high severity alerts rather than urgent severity alerts. Defaults to 30m.
/alertmanager/notifications/high_repeat_interval	Same as *urgent_repeat_interval*, except for high severity alerts rather than urgent severity alerts. Defaults to 4h.
/alertmanager/notifications/medium_group_wait	Same as *urgent_group_wait*, except for medium severity alerts rather than urgent severity alerts. Defaults to 10m.
/alertmanager/notifications/medium_group_interval	Same as *urgent_group_interval*, except for medium severity alerts rather than urgent severity alerts. Defaults to 1h.

General Notification Setting	Description
/alertmanager/notifications/medium_repeat_interval	Same as *urgent_repeat_interval*, except for medium severity alerts rather than urgent severity alerts. Defaults to 24h.
/alertmanager/notifications/low_group_wait	Same as *urgent_group_wait*, except for low severity alerts rather than urgent severity alerts. Defaults to 1h.
/alertmanager/notifications/low_group_interval	Same as *urgent_group_interval*, except for low severity alerts rather than urgent severity alerts. Defaults to 12h.
/alertmanager/notifications/low_repeat_interval	Same as *urgent_repeat_interval*, except for low severity alerts rather than urgent severity alerts. Defaults to 7d.

3.3.4.5. Email Notification Settings

To configure HyperIQ to send alert notifications to an email address, use *iqmgr config set* commands to assign values to some or all of the settings listed below. Remember to subsequently run *iqmgr config apply*.

Email Notification Setting	Description
/alertmanager/notifications/email/smarthost	Required. SMTP host through which emails are sent. Include the port number, for example *smtp.example.org:587* If you specify 465 as the SMTP host's listening port, then HyperIQ's SMTP client assumes that Implicit TLS is being used -- not STARTTLS (Explicit TLS).
/alertmanager/notifications/email/to	Required. Email address to which to send notifications.
/alertmanager/notifications/email/from	Required. Email address to use in the From field of email notifications.
/alertmanager/notifications/email/auth_username	Optional. SMTP authentication username.
/alertmanager/notifications/email/auth_password	Optional. SMTP authentication password, for SMTP Auth using LOGIN and PLAIN. **Note** For the authentication password setting do not include the password value on the command line. Instead just run *iqmgr config set /alertmanager/notifications/email/auth_password* and then you will be prompted to provide the password.
/alertmanager/notifications/email/auth_secret	Optional. SMTP authentication secret, for SMTP Auth using CRAM-MD5.

Email Notification Setting	Description
	Note For the authentication secret setting do not include the secret value on the command line. Instead just run *iqmgr config set /alert-manager/notifications/email/auth_secret* and then you will be prompted to provide the secret.
/alertmanager/notifications/email/insecure_skip_verify	Optional, defaults to false. If set to *true*, the HyperIQ SMTP client will not validate the SMTP server's TLS/SSL certificate. Set this to *true* if your SMTP host uses a self-signed certificate.
/alertmanager/notifications/email/require_tls	Optional, defaults to true. If set to *true* -- as it is by default -- SMARTTLS will be used and required, unless the SMTP host port is 465 (if the port is 465, Implicit TLS is used instead). Set to *false* if you want to allow non-encrypted connections between the HyperIQ SMTP client and the SMTP host, on a port other than 465.
/alertmanager/notifications/send_resolved	Optional, defaults to false. If set to *true*, HyperIQ will send an email notification when an alert condition is resolved (for example, when a service that has been down comes back up again).
/alertmanager/notifications/urgent_group_wait	Optional, defaults to 10s. How long to initially wait to send a notification for a group of similar urgent alerts. Allows time to wait for an inhibiting alert to arrive, or to collect more initial alerts for the same group before sending the notification. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/urgent_group_interval	Optional, defaults to 30s. How long to wait before sending a notification about new urgent alerts that are added to a group of urgent alerts for which an initial notification has already been sent. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/urgent_repeat_interval	Optional, defaults to 30m. How long to wait before sending a notification again if it has already been sent successfully for an urgent

Email Notification Setting	Description
	alert. Supported value formats are *<n>s* (for number of seconds) or *<n>m* (number of minutes) or *<n>h* (number of hours).
/alertmanager/notifications/high_group_wait	Same as *urgent_group_wait*, except for high severity alerts rather than urgent severity alerts. Defaults to 5m.
/alertmanager/notifications/high_group_interval	Same as *urgent_group_interval*, except for high severity alerts rather than urgent severity alerts. Defaults to 30m.
/alertmanager/notifications/high_repeat_interval	Same as *urgent_repeat_interval*, except for high severity alerts rather than urgent severity alerts. Defaults to 4h.
/alertmanager/notifications/medium_group_wait	Same as *urgent_group_wait*, except for medium severity alerts rather than urgent severity alerts. Defaults to 10m.
/alertmanager/notifications/medium_group_interval	Same as *urgent_group_interval*, except for medium severity alerts rather than urgent severity alerts. Defaults to 1h.
/alertmanager/notifications/medium_repeat_interval	Same as *urgent_repeat_interval*, except for medium severity alerts rather than urgent severity alerts. Defaults to 24h.
/alertmanager/notifications/low_group_wait	Same as *urgent_group_wait*, except for low severity alerts rather than urgent severity alerts. Defaults to 1h.
/alertmanager/notifications/low_group_interval	Same as *urgent_group_interval*, except for low severity alerts rather than urgent severity alerts. Defaults to 12h.
/alertmanager/notifications/low_repeat_interval	Same as *urgent_repeat_interval*, except for low severity alerts rather than urgent severity alerts. Defaults to 7d.

For example:

```
iqmgr config set /alertmanager/notifications/email/smarthost smtp.enterprise.com:465
iqmgr config set /alertmanager/notifications/email/to administrator@enterprise.com
iqmgr config set /alertmanager/notifications/email/from hyperiq@enterprise.com
iqmgr config apply
```

After configuring and applying your email notification settings you can send a **test alert notification email** by running this command:

```
iqmgr alerts test
```

The alert name in this test notification email will be "dummy_alert".

3.3.4.6. Slack Notification Settings

To configure HyperIQ to send alert notifications to Slack, use *iqmgr config set* commands to assign values to **both** of the settings listed below. Remember to subsequently run *iqmgr config apply*.

Slack Notification Settings	Description
/alertmanager/notifications/slack/api_url	Required. Slack webhook URL. Include the protocol prefix ("https://") in the URL value.
/alertmanager/notifications/slack/channel	Required. Slack channel or user to which to send notifications.

3.3.4.7. MS Teams Notification Setting

To configure HyperIQ to send alert notifications to Microsoft Teams, use *iqmgr config set* commands to assign a value to the setting below. Remember to subsequently run *iqmgr config apply*.

> **Note** Before enabling this feature in HyperIQ, configure a new "Incoming webhook" in your MS Teams application (under Channel -> Connectors), and make a note of the webhook URL.

MS Teams Notification Setting	Description
/alertmanager/notifications/msteams_webhook	Required. MS Teams webhook URL. **Exclude** the protocol prefix ("https://") from the URL value.

3.3.4.8. Splunk Notification Settings

To configure HyperIQ to send alert notifications to Splunk, use *iqmgr config set* commands to assign values to **both** of the settings listed below. Remember to subsequently run *iqmgr config apply*.

> **Note** Before enabling this feature in HyperIQ, in your Splunk application create a new HTTP Event Collector (receiver) and a new alert, making notes of the receiver's token value and the alert's webhook URL. For an example of configuring Splunk to receive alerts from HyperIQ, see the addendum titled "Sending alerts to Splunk" at the end of this document.

Splunk Notification Settings	Description
/alertmanager/notifications/splunk_url	Required. Splunk alert webhook URL. Include the protocol prefix ("https://") in the URL value.
/alertmanager/notifications/splunk_token	Required. Splunk HTTP Event Collector token.
/alertmanager/notifications/splunk_port	Optional. Splunk listening port. Defaults to 8088.

3.3.5. Alert Codes

Each HyperIQ alert has a unique alert code. For alerts that may appear both in HyperIQ and in HyperStore's Cloudian Management Console (CMC), the same alert code is used by both applications. For example, if the S3 Service is down on a node, the alert in the CMC and the alert in HyperIQ both have the alert code "SS000007".

Prometheus Alerts

- PrometheusTargetMissing -- PR000001
- PrometheusAllTargetsMissing -- PR000002
- PrometheusConfigurationReloadFailure -- PR000003
- PrometheusTooManyRestarts -- PR000004
- PrometheusAlertmanagerConfigurationReloadFailure -- AM000001
- PrometheusRuleEvaluationSlow -- PR000005
- PrometheusNotificationsDropped -- PR000006
- PrometheusAlertmanagerNotificationFailing -- AM000002

Service Alerts

- ServiceDown:
  - "cassandra" -- SS000002
  - "hyperstore" -- SS000003
  - "cloudians3" -- SS000007
  - "admin" -- SS000001
  - "redismon" -- SS000006
  - "redis-credentials" -- SS000005
  - "redis-qos" -- SS000004
  - "iq_s3" -- SA000003
  - "iq_node" -- SA000004
  - "iq_service" -- SA000005
  - "iq_admin" -- SA000006
  - "iq_jmx" -- SA000007
  - "ntpd" -- SA000008
- CassandraExceptions -- SA000001
- EnterpriseLicenseNotInstalled -- SA000002

Node Alerts

- NodeDown -- GS000001
- NodeOutOfMemory -- ND000001
- NodeHighInboundNetworkTraffic -- NS000007
- NodeHighOutboundNetworkTraffic -- NS000008
- NodeOutOfDiskSpace -- GS000003
- NodeOutOfInodesHigh -- ND000002
- NodeOutOfInodesCritical -- ND000003
- HighCPUUsage -- GS000006
- NodeRAIDDiskFailure -- ND000004
- NodeKernelVersionDeviations -- ND000005
- NodeHyperStoreVersionDeviations -- ND000006
- NodeNetworkInterfaceFlapping -- ND000007
- NodeInterfaceDown -- ND000008

- NodeHighTCPInErrs -- ND000009
- NodeHighTCPRetransHigh -- ND000010
- NodeNetworkBondDegraded -- ND000011
- NodeTimeOutOfSyncWithHIQ -- ND000012
- NodeTimeNotSyncedWithNTP -- ND000013
- NodePSUNotActive -- ND000014
- NodeDiskWillFillIn30Days -- DS000001
- NodeUnusualDiskReadLatency -- DS000002
- NodeUnusualDiskWriteLatency -- DS000003
- NodeDiskFailure -- GS000005
- DataDiskOutOfSpace -- GS000004
- RootDiskOutOfSpace -- DS000004
- DiskSMARTHealthStatus -- DS000005

S3 Analytics Alerts (**Disabled by default** except for "S3LogLinesIgnored")

- S3LogLinesIgnored -- S3000001
- User30DayDownloadLimitExceeded -- S3000002
- User30DayUploadLimitExceeded -- S3000003
- GETTransactionsPerSecond -- NS000001
- PUTTransactionsPerSecond -- NS000002
- DELETETransactionsPerHour -- S3000004
- TotalSystemBytecountLimitExceeded -- S3000005
- TotalSystemObjectLimitExceeded -- S3000006
- GETThroughputLimitExceeded -- NS000003
- PUTThroughputLimitExceeded -- NS000004
- 503sLast10MinLimitExceeded -- S3000007

3.4. Configuring LDAP Authentication for HyperIQ Users

To configure HyperIQ to support LDAP authentication when a user logs into the HyperIQ UI, use *iqmgr config set* commands to assign values to the settings listed below. Remember to subsequently run *iqmgr config apply*. (For an overview of *iqmgr config* command usage, see the introduction to **"Configuring Alerts and Notifications"** (page 37)).

When LDAP authentication is configured, if a new user logs into the HyperIQ UI with their LDAP-based login credentials and if HyperIQ successfully authenticates the user with your LDAP system, HyperIQ automatically creates a HyperIQ user account for the user (and then a HyperIQ administrator can see the user's account in the list of users, and optionally adjust the user's HyperIQ role and profile information). Each time the user subsequently logs into HyperIQ, HyperIQ will check with your LDAP system to validate the user's login credentials.

LDAP Setting	Description
/grafana/ldap/enabled	Enable LDAP for HyperIQ -- true or false. Defaults to false.
/grafana/ldap/allow_sign_up	Allow new LDAP-authenticated Grafana users to be created -- true or false. Defaults to true.

LDAP Setting	Description
/grafana/ldap/server_host	Required if LDAP is enabled. LDAP server host
/grafana/ldap/search_base_dns	Required if LDAP is enabled. Array of base DNs to search, e.g. '["dc=your_domain,dc=com"]'
/grafana/ldap/bind_dn	Required if LDAP is enabled. Search user bind DN template, e.g. '%s@your_domain.com'
/grafana/ldap/bind_password	Search user bind password. If the password contains # or *;* you have to wrap it with triple quotes, e.g. """#password;""" **Note** * For the bind password setting do not include the password value on the command line. Instead just run *iqmgr config set /grafana/ldap/bind_password* and then you will be prompted to provide the bind password. * Setting *bind_password* can be skipped if you are able to set a *bind_dn* that matches all possible users, for example ""cn=%s,o=users,dc=grafana,dc=org". For more information on Grafana LDAP settings see **https://grafana.com/docs/grafana/latest/setup-grafana/configure-security/configure-authentication/ldap/**.
/grafana/ldap/server_port	LDAP server port. Defaults to 389 (or to 636 if 'use_ssl' is set to 'true')
/grafana/ldap/search_filter	User search filter, default: '(sAMAccountName=%s)'
/grafana/ldap/use_ssl	Set to true if connections to the LDAP server should use an encrypted TLS connection (either with STARTTLS or LDAPS). Defaults to false.
/grafana/ldap/start_tls	If set to true, use LDAP with STARTTLS instead of LDAPS. Defaults to false.
/grafana/ldap/client_cert	If using LDAPS for connecting to LDAP server: Path in *hyperiq_grafana* container to your client certificate (for example */home/my.crt*) **Note** For this and for the client private key and root CA certificate settings below, you must copy the certificate and key files into the *hyperiq_grafana* container, so that the files are at the location(s) that you specify with these settings. You can copy files into the container with: *docker cp <local-path-to-file> <container>:<destination directory>* or *iqmgr cp <local-path-to-file> <container>:<destination*

LDAP Setting	Description
	directory> For example: *docker cp ./ca.crt hyperiq_grafana:/home/grafana/* For more on *iqmgr cp* see **"Using the iqmgr Tool to Manage HyperIQ"** (page 27).
/grafana/ldap/client_key	If using LDAPS for connecting to LDAP server: Path in *hyperiq_grafana* container to your client private key (for example */home/my.key*)
/grafana/ldap/root_ca_cert	If using LDAPS for connecting to LDAP server: Path in *hyperiq_grafana* container to your root CA certificate (for example */home/ca.crt*)
/grafana/ldap/ssl_skip_verify	If using LDAPS for connecting to LDAP server: Set to true if you want to skip SSL certificate validation (relevant only if "use_ssl" is set to true). Defaults to false.

3.5. Uninstalling HyperIQ

Uninstalling HyperIQ will stop and remove Docker containers, images, networks and volumes, and the HyperIQ logs and tools directory. **Your HyperIQ data will be lost** (you will lose accumulated HyperStore monitoring data stored in the HyperIQ database, from the period during which you've had HyperIQ running).

To uninstall HyperIQ, log into the HyperIQ host and run the appropriate command as indicated below.

> **Note** By default, the uninstall operation will also remove the Prometheus metrics exporters from all your HyperStore nodes (these exporters were installed on your HyperStore nodes during the HyperIQ setup process). If you want to leave these exporters in place on your HyperStore nodes, use the ***--keep-exporters*** option.

HyperIQ virtual appliance:

```
iquninstall [--keep-exporters]
```

HyperIQ installed in an existing Docker environment on Linux:

Change into the directory in which you downloaded the *hyperiq-1.9.bin* file before you installed HyperIQ, and then run:

```
./hyperiq-1.9.bin uninstall [--keep-exporters]
```